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OPTICS



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The Propagation of Light

4

4.1 INTRODUCTION

We now consider a number of phenomena related to the propagation of light and its interaction with material media. In particular, we shall study the characteristics of light waves as they progress through various substances, crossing interfaces, and being reflected and refracted in the process. For the most part, we shall envision light as a classical electromagnetic wave whose velocity through any medium is dependent upon that material's electric and magnetic properties. It is an intriguing fact that many of the basic principles of optics are predicated on the wave aspects of light and are yet completely independent of the exact nature of that wave. As we shall see, this accounts for the longevity of *Huygens' principle* which has served in turn to describe mechanical ether waves, electromagnetic waves, and now after three hundred years applies to quantum optics.

Suppose, for the moment, that a wave impinges on the interface separating two different media (e.g. a piece of glass in air). As we know from our everyday experiences, a portion of the incident flux density will be diverted back in the form of a *reflected wave*, while the remainder will be transmitted across the boundary as a *refracted wave*. On a submicroscopic scale we envisage an assemblage of atoms which scatter the incident radiant energy. The manner in which these emitted light wavelets superimpose and combine with each other will depend on the spatial distribution of the scattering atoms. As we know from the previous chapter, the scattering process is responsible for the *index of refraction* as well as the resultant *reflected* and *refracted* waves. This atomistic description is quite satisfying conceptually even though it is not a simple matter to treat analytically. It should, however, be kept in mind even when applying macroscopic techniques, as indeed we shall, later on.

We now seek to determine the general principles governing or at least describing the propagation, reflection and refraction of light. In principle it should be possible to trace the progress of radiant energy through any system by applying Maxwell's equations and the associated boundary conditions. In practice, however, this is often an impractical if not an impossible task (see Section 10.1). And so we shall take a somewhat different route, stopping, when appropriate, to verify that our results are in accord with electromagnetic theory.

4.2 THE LAWS OF REFLECTION AND REFRACTION

4.2.1 Huygens' Principle

Recall that a wavefront is a surface over which an optical disturbance has a constant phase. As an illustration, Fig. 4.1 shows a small portion of a spherical wavefront Σ emanating from a monochromatic point source S in a homogeneous medium. Clearly if the radius of the wavefront as shown is r, at some later time t it will simply be $(r + vt)$ where v is the phase velocity of the wave. But suppose instead that the light passes through a nonuniform sheet of glass as in Fig. 4.2 so that the wavefront itself is distorted. How can we determine its new form Σ'? Or for that matter, what will Σ' look like at some later time if it is allowed thereafter to continue unobstructed?

A preliminary step toward the solution of this problem appeared in print in 1690 in the work entitled *Traité de la Lumière* which had been written twelve years earlier by the Dutch physicist Christiaan Huygens. It was there that he enunciated what has since become known as *Huygens' principle*, that *every point on a primary wavefront serves as*

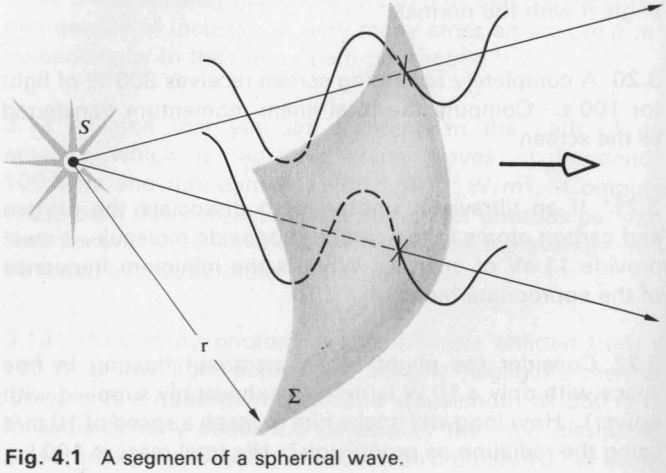


Fig. 4.1 A segment of a spherical wave.

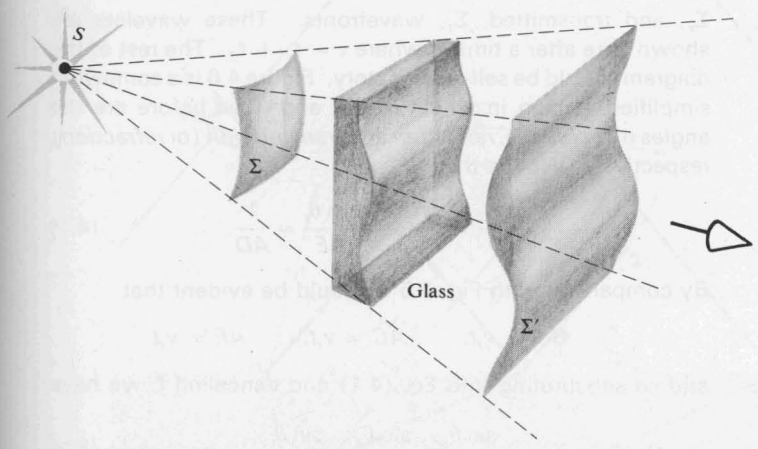


Fig. 4.2 Distortion of a portion of a wavefront on passing through a material of nonuniform thickness.

the source of spherical secondary wavelets such that the primary wavefront at some later time is the envelope of these wavelets. Moreover, the wavelets advance with a speed and frequency equal to that of the primary wave at each point in space. If the medium is homogeneous the wavelets may be constructed with finite radii, whereas if it is inhomogeneous the wavelets will have to have infinitesimal radii. Figure 4.3 should make all of this fairly clear; it shows a view of a wavefront Σ as well as a number of spherical secondary wavelets which, after a time t, have propagated out to a radius of vt. The envelope of all of these wavelets is then asserted to correspond to the advanced primary wave Σ'. It is easy to visualize the process in terms of mechanical vibrations of an elastic medium. Indeed this is the way that Huygens envisioned it within the context of an all-pervading ether as is evident from this comment by him:

> We have still to consider, in studying the spreading out of these waves, that each particle of matter in which a wave proceeds not only communicates its motion to the next particle to it, which is on the straight line drawn from the luminous point, but that it also necessarily gives a motion to all the others which touch it and which oppose its motion. The result is that around each particle there arises a wave of which this particle is a center.

We can make use of these ideas on two different levels: In treating diffraction theory, a mathematical representation of the wavelets will serve as the basis for a valuable analytical technique. Using it, one is able to trace the progress of a primary wave past all sorts of apertures and obstacles by summing up the wavelet contributions mathematically. In contrast, Fig. 4.3 in itself represents a graphical application of the essential ideas and as such is known as *Huygens' construction*. For the moment we limit ourselves to the latter approach.

Thus far we have merely stated Huygens' principle without any justification or proof of its validity. As we shall see (Chapter 10) Fresnel in the eighteen hundreds successfully modified Huygens' principle somewhat. A little later on Kirchhoff showed that the *Huygens–Fresnel principle* was a direct consequence of the differential wave equation (2.59), thereby putting it on a firm mathematical base. That there was a need for a bit of reformulation of the principle is evident from Fig. 4.3 where we deceptively only drew hemispherical wavelets. Had we drawn them as spheres, there would have been a *back wave* moving toward the source—something which is not observed. Since this difficulty was taken care of theoretically by Fresnel and Kirchhoff, we shall not be disturbed by it. In fact, we shall merely overlook it completely when applying Huygens' construction.

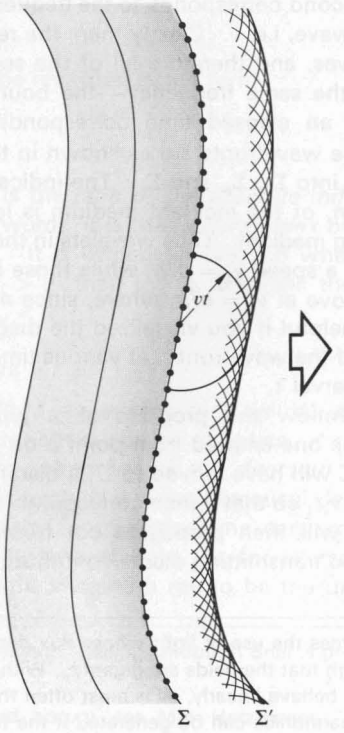


Fig. 4.3 The propagation of a wavefront via Huygens' principle.

Huygens' principle fits in rather nicely with our earlier discussion of the atomic scattering of radiant energy. Each atom of a material substance which interacts with an incident primary wavefront can be regarded as a point source of scattered secondary wavelets. Things are not quite as clear when we consider the principle as applied to the propagation of light through a vacuum. It is helpful however to keep in mind that at any point in empty space on the primary wavefront there exists both a time-varying **E**-field and a time-varying **B**-field. Both of these in turn create new fields which move out from that point. In this sense each point on the wavefront is analogous to a physical scattering center.

4.2.2 Snell's Law and the Law of Reflection

Imagine that we have a monochromatic plane wave incident on the smooth interface separating two different transparent media as in Fig. 4.4. We now wish to determine the wave's behavior using Huygens' construction. Accordingly, each point of the plane wavefront Σ in Fig. 4.5 will be taken as the source of a secondary wavelet.

The number of primary wavefronts arriving at the interface per second corresponds to the frequency of the incident harmonic wave, i.e. ν. Clearly then, the reflected and transmitted waves, and therefore all of the secondary wavelets, will have the same frequency—the boundary cannot alter ν.* After an elapsed time corresponding to the period $\tau = 1/\nu$, the wavefronts are as shown in the diagram where Σ has split into Σ_i, Σ_t, and Σ_r. The indices of refraction are such that n_i of the incident medium is less than n_t of the transmitting medium. Thus wavelets in the incident medium travel with a speed $v_i = c/n_i$ while those in the transmitting medium move at $v_t = c/n_t$ where, since $n_t > n_i$, $v_t < v_i$. It might be helpful if you visualized the diagram as a multiple exposure of the wavefront Σ at various times and particularly after an interval τ.

Let's follow the progress of a typical wavelet, for example the one emitted from point b on Σ. After a time t_1 the plane Σ will have moved to Σ', a distance in the incident medium of $v_i t_1$ so that b then corresponds to b'. Presumably, a wavelet will then propagate out from b' into both the incident and transmitting media contributing to the reflected, Σ_r, and transmitted, Σ_t, wavefronts. These wavelets are shown here after a time t_2 where $\tau = t_1 + t_2$. The rest of the diagram should be self-explanatory. Figure 4.6 is a somewhat simplified version in which θ_i, θ_r, and θ_t as before are the angles of *incidence*, *reflection* and *transmission* (or *refraction*) respectively. Notice that

$$\frac{\sin\theta_i}{BD} = \frac{\sin\theta_r}{AC} = \frac{\sin\theta_t}{AE} = \frac{1}{AD}. \tag{4.1}$$

By comparison with Fig. 4.5 it should be evident that

$$BD = v_i t, \qquad AC = v_i t, \qquad AE = v_t t$$

and so substituting into Eq. (4.1) and canceling t, we have

$$\frac{\sin\theta_i}{v_i} = \frac{\sin\theta_r}{v_i} = \frac{\sin\theta_t}{v_t}. \tag{4.2}$$

It then follows from the first two terms that *the angle of incidence equals the angle of reflection*, that is

$$\theta_i = \theta_r. \tag{4.3}$$

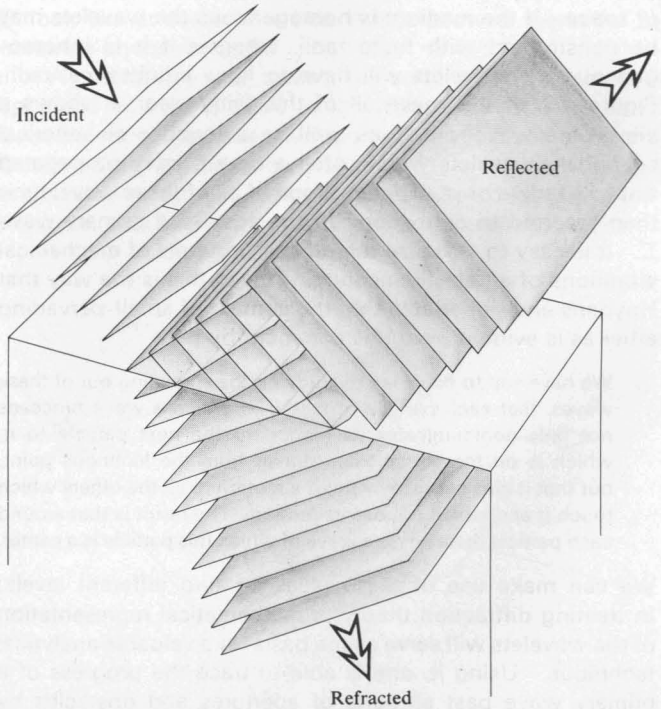


Fig. 4.4 Reflection and transmission of plane waves.

* This presumes the use of light whose flux density is not so extraordinarily high that the fields are gigantic. With this assumption the medium will behave linearly, as is most often the case. In contrast, observable harmonics can be generated if the fields are made large enough (Section 14.4).

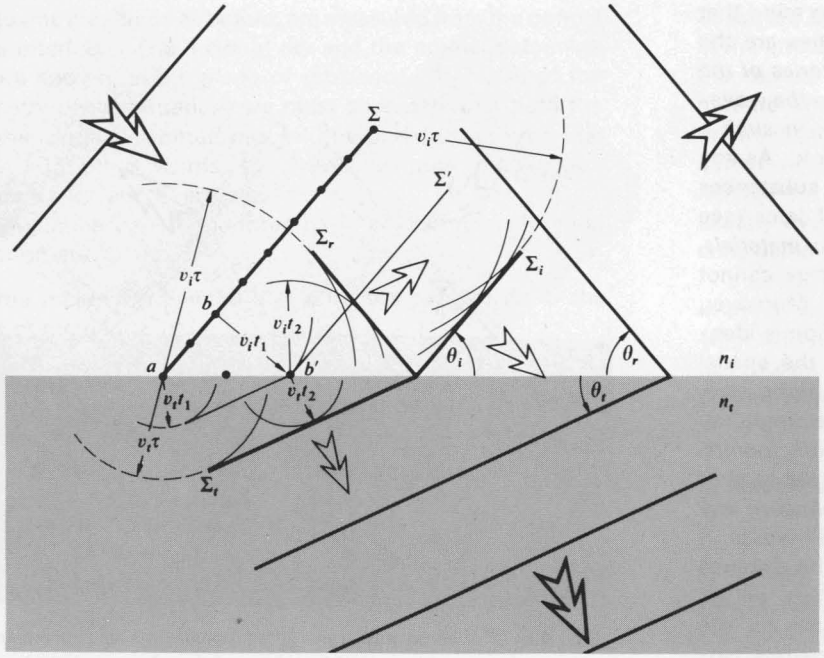


Fig. 4.5 Reflection and transmission at an interface via Huygens' principle.

Known as the *law of reflection*, it first appeared in the book entitled *Catoptrics* which was purported to have been written by Euclid.

The first and last terms of Eq. (4.2) yield

$$\frac{\sin\theta_i}{\sin\theta_t} = \frac{v_i}{v_t} \tag{4.4}$$

or since $v_i/v_t = n_t/n_i$

$$n_i \sin\theta_i = n_t \sin\theta_t. \tag{4.5}$$

This is the very important *law of refraction*, the physical consequences of which have been studied, at least on record, for over eighteen hundred years. On the basis of some very fine observations Claudius Ptolemy of Alexandria attempted unsuccessfully to divine the expression. Kepler very nearly succeeded in deriving the law of refraction in his book *Supplements to Vitello* in 1604. Unfortunately he was misled by some erroneous data compiled earlier by Vitello (ca. 1270). The correct relationship seems to have been arrived at independently by Snell* at the University of Leyden and the French mathematician Descartes.† In English-speaking countries Eq. (4.5) is generally referred to as *Snell's law*.

Notice that it can be rewritten in the form

$$\frac{\sin\theta_i}{\sin\theta_t} = n_{ti} \tag{4.6}$$

where $n_{ti} \equiv n_t/n_i$ is the *ratio of the absolute indices of refraction*. In other words, it is the *relative index of refraction of the two media*. It is evident in Fig. 4.5 where $n_{ti} > 1$, i.e. $n_t > n_i$ and $v_i > v_t$ that $\lambda_i > \lambda_t$ whereas the opposite would be true if $n_{ti} < 1$.

4.2.3 Light Rays

The concept of a light ray is one which will be of sustained interest to us throughout our study of optics. *A ray is a line drawn in space corresponding to the direction of flow of radiant energy*. As such, it is a mathematical device rather than a physical entity. In practice one can produce very narrow *beams* or *pencils* of light (as for example a laser beam) and we might imagine a ray to be the unattainable

* This is the most common spelling, although Snel is probably more accurate.

† For a more detailed history, see Max Herzberger, "Optics from Euclid to Huygens," *Appl. Opt.* **5**, 1383 (1966).

limit on the narrowness of such a beam. Bear in mind that in an *isotropic medium*, i.e. one whose properties are the same in all directions, *rays are orthogonal trajectories of the wavefronts*. That is to say, *they are lines normal to the wavefronts at every point of intersection*. Evidently, *in such a medium a ray is parallel to the propagation vector* **k**. As you might suspect, this is not true in *anisotropic* substances whose further consideration we postpone until later (see Section 8.4.1). *Within homogeneous isotropic materials, rays will be straight lines* since by symmetry they cannot bend in any preferred direction, there being none. Moreover, as a result of the fact that the speed of propagation is identical in all directions within a given medium, the spatial separation between two wavefronts, measured along rays, must be the same everywhere.* Points where a single ray intersects a set of wavefronts are called *corresponding points*, as for example A, A', and A'' in Fig. 4.7. *Evidently the separation in time between any two corresponding points on any two sequential wavefronts is identical.* In other words, if wavefront Σ transforms into Σ'' after a time t'', the distance between corresponding points on any and all rays will be traversed in that same time t''. This will obviously be true even if the wavefronts pass from one homogeneous isotropic medium into another. This just means that each point on Σ can be imagined as following the path of a ray to arrive at Σ'' in the time t''.

If a group of rays is such that we can find a surface which is orthogonal to each and every one of them, they are said to form a *normal congruence*. For example, the rays emanating from a point source are perpendicular to a sphere centered at the source and consequently form a normal congruence.

We can now briefly consider an alternative technique to Huygens' principle which will also allow us to follow the progress of light through various isotropic media. The basis for this approach is the *theorem of Malus and Dupin* (introduced in 1808 by E. Malus and modified in 1816 by C. Dupin) according to which *a group of rays will preserve its normal congruence after any number of reflections and refractions* (as in Fig. 4.7). Looking at it from our present vantage point of the wave theory, it is evidently equivalent to the statement that rays remain orthogonal to wavefronts throughout all propagation processes in isotropic media. As shown in Problem 4.9, the theorem can be used to derive the law of reflection as well as Snell's law. It is often most convenient to carry out a ray trace through an optical system using the laws of reflection and refraction and then reconstruct the wavefronts. The latter can be accomplished in accord with the above considerations of equal transit times between corresponding points and the orthogonality of the rays and wavefronts.

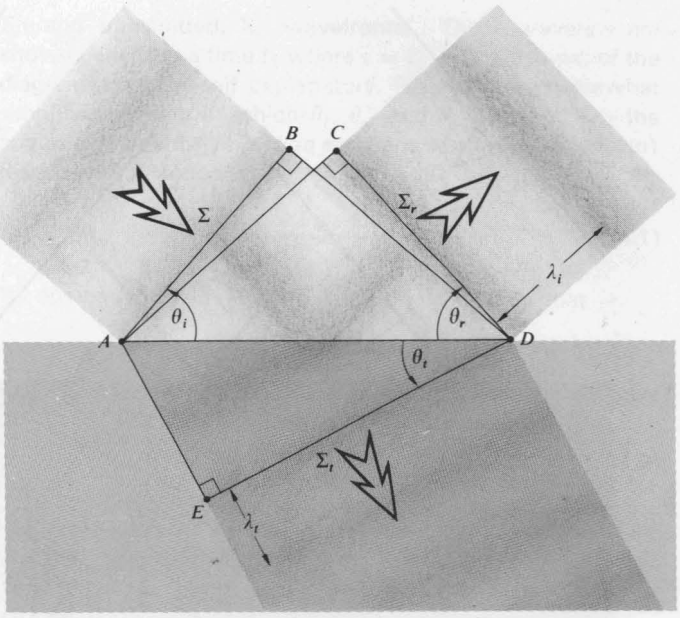


Fig. 4.6 Reflected and transmitted wavefronts at a given instant.

Figure 4.8 depicts the parallel ray formation concomitant with a plane wave where now θ_i, θ_r, and θ_t, which have the

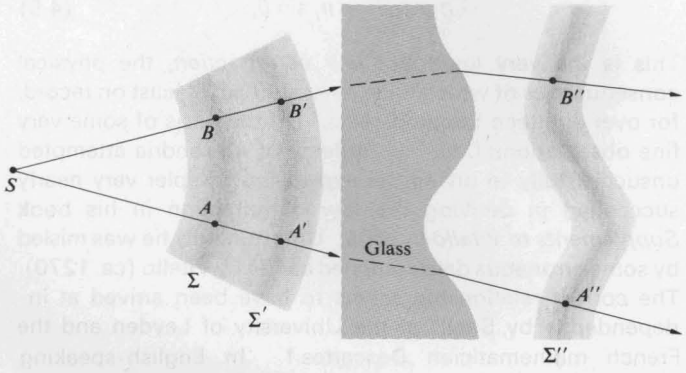


Fig. 4.7 Wavefronts and rays.

* When the material is inhomogeneous or when there is more than one medium involved, it will be the *optical path length* (see Section 4.2.4) between the two wavefronts which is the same.

exact same meanings as before, are measured from the normal to the interface. The incident ray and the normal determine a plane known as the *plane of incidence*. Because of the symmetry of the situation, we must anticipate that both the reflected and transmitted rays will be undeflected from that plane. In other words, the respective unit propagation vectors $\hat{\mathbf{k}}_i$, $\hat{\mathbf{k}}_r$, and $\hat{\mathbf{k}}_t$ are coplanar.

In summary then, the three basic laws of reflection and refraction are:

1. the incident, reflected and refracted rays all lie in the plane of incidence.
2. $\theta_i = \theta_r$ [4.3]
3. $n_i \sin \theta_i = n_t \sin \theta_t$. [4.5]

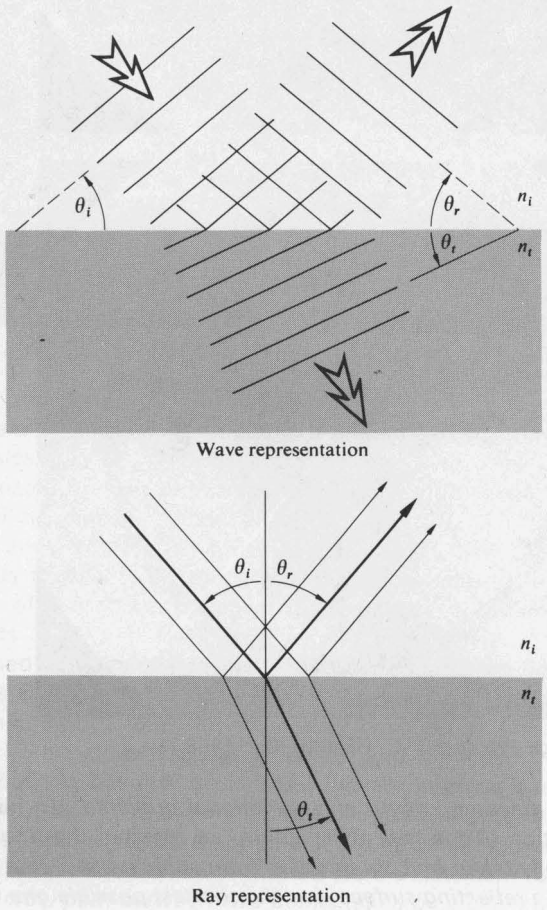


Fig. 4.8 The wave and ray representations of an incident, reflected and transmitted beam.

These are illustrated rather nicely using a narrow light beam in the photographs of Fig. 4.9. In this case as in Fig. 4.10(a) the surface is smooth, or more precisely, any irregularities in it are small compared to a wavelength. A sharply defined or *specularly reflected* beam emerges. In contrast, the *diffuse reflection* of Fig. 4.10(b) occurs when the surface is relatively rough. Despite this, the law of reflection holds exactly over any region which is small enough so as to be considered smooth.

Let $\hat{\mathbf{u}}_n$ be a unit vector normal to the interface pointing in the direction from the incident to the transmitting medium (Fig. 4.11). It can be shown (as you will hopefully prove in Problem 4.10) that the first and third basic laws may be combined in the form of a *vector refraction equation*

$$n_i(\hat{\mathbf{k}}_i \times \hat{\mathbf{u}}_n) = n_t(\hat{\mathbf{k}}_t \times \hat{\mathbf{u}}_n) \tag{4.7}$$

or alternatively

$$n_t\hat{\mathbf{k}}_t - n_i\hat{\mathbf{k}}_i = (n_t \cos \theta_t - n_i \cos \theta_i)\hat{\mathbf{u}}_n. \tag{4.8}$$

It is a bit premature at this point to discuss in detail the processes which on an atomic scale result specifically in the laws of reflection and refraction. As we shall see, scattered wavelets emitted by atoms in the material superimpose, generating a reflected and refracted beam only in the directions of $\hat{\mathbf{k}}_r$ and $\hat{\mathbf{k}}_t$. Reflection is obviously a surface effect, in fact, it involves only those atoms in a layer about $\lambda/2$ deep. The response is very much like a transmitting antenna array where the spacing (in this case $\sim 10^{-10}$ m) between sources is less than λ (see Section 10.1.3). In contrast, if the incident wavelength is reduced below that of light (~ 500 nm) to the x-ray region ($\sim 10^{-10}$ m) which is of the order of the atomic spacing, there will generally no longer be flux density maxima simply in the directions of θ_r and θ_t as in the case of light. Similarly, if we use some sort of arrangement where the scattering centers have a separation of the order of λ for light (e.g. a diffraction grating), we can observe higher-order reflected and transmitted beams in addition to those (zeroth-order ones) considered above (see Fig. 10.36).

4.2.4 Fermat's Principle

The laws of reflection and refraction and indeed the manner in which light propagates in general, can be viewed from yet another entirely different and intriguing perspective afforded us by something called *Fermat's principle*. The ideas which will unfold presently have had a tremendous influence on the development of physical thought in and beyond the study

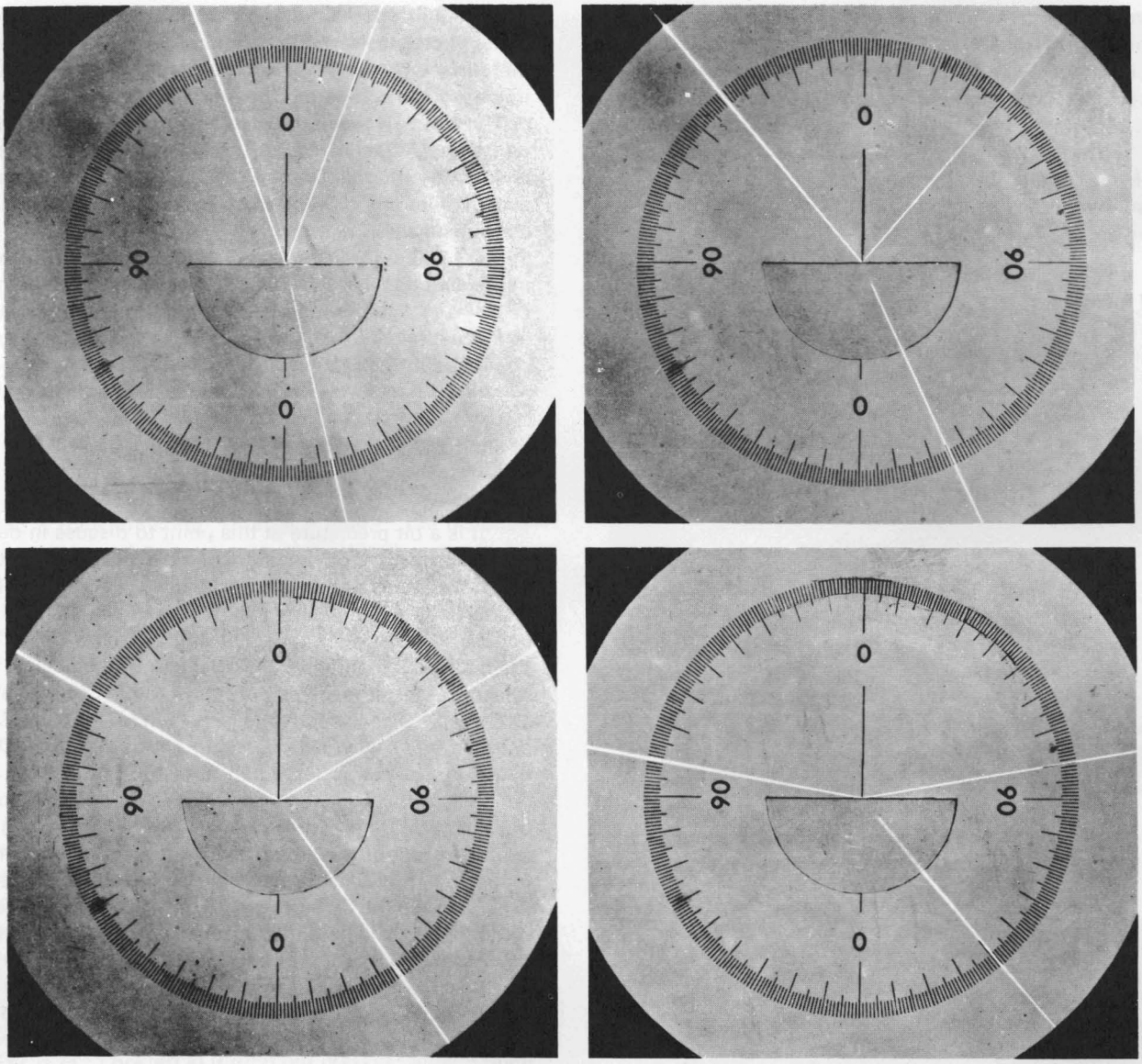
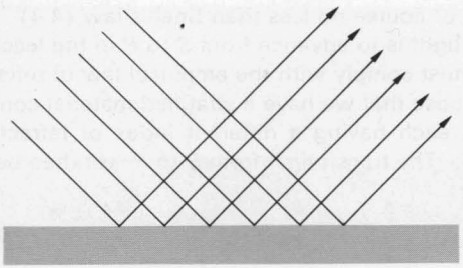


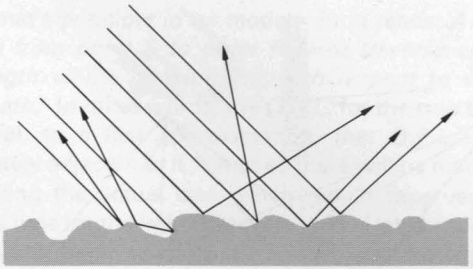
Fig. 4.9 Refraction at various angles of incidence. [Photos courtesy *PSSC College Physics*, D. C. Heath & Co., 1968.]

of classical optics. Apart from its implications in quantum optics (Section 13.6, p. 453) Fermat's principle provides us with an insightful and highly useful way of appreciating and anticipating the behavior of light.

Hero of Alexandria, who lived some time between 150 B.C. and 250 A.D., was the first to set forth what has since become known as a *variational principle*. In his formulation of the law of reflection, he asserted that *the path actually taken by light in going from some point S to a point P via a reflecting surface was the shortest possible one*. This can be seen rather easily in Fig. 4.12 which depicts a point source *S* emitting a number of rays which are then "reflected"



Specular



Diffuse

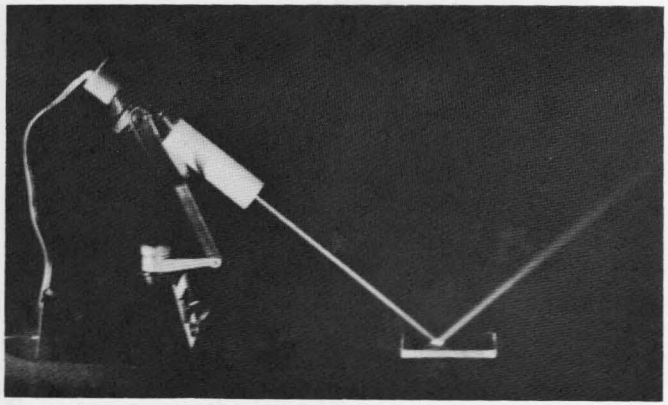


Fig. 4.10 (a) Specular reflection.

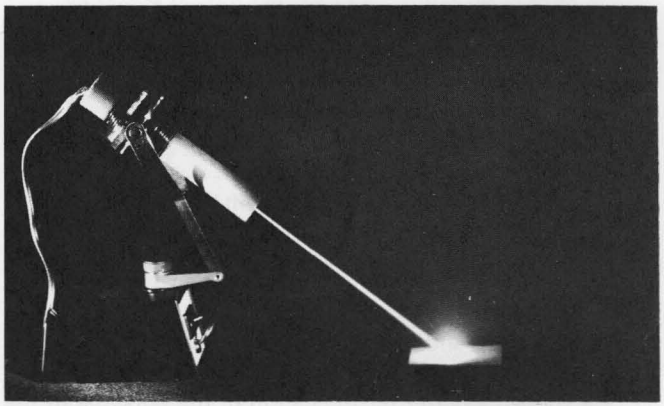


Fig. 4.10 (b) Diffuse reflection. [Photos courtesy *PSSC College Physics*, D. C. Heath & Co., 1968.]

toward P. Of course only one of these paths will have any physical reality. If we simply draw the rays as if they emanated from S' (the image of S), none of the distances to P will have been altered, i.e. $SAP = S'AP$, $SBP = S'BP$, etc. But obviously the straight-line path $S'BP$, which corresponds to $\theta_i = \theta_r$, is the shortest possible one. The same kind of reasoning (Problem 4.13) makes it evident that points S, B, and P must lie in what has previously been defined as the plane of incidence. For over fifteen hundred years Hero's curious observation stood alone until in 1657 Fermat propounded his celebrated *principle of least time* which encompassed both reflection and refraction. Quite obviously a beam of light traversing an interface does not take a straight line or *minimum spatial path* between a point in the incident medium and one in the transmitting medium. Fermat consequently reformulated Hero's statement to read: *the actual path between two points taken by a beam of light is the one which is traversed in the least time.* As we shall see in a little while even this form of the statement is some-

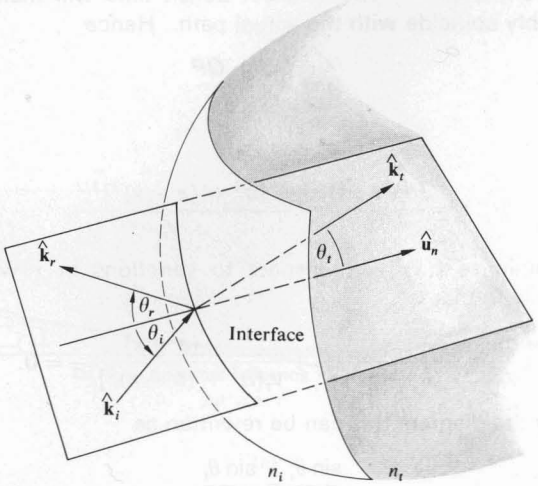


Fig. 4.11 The ray geometry.

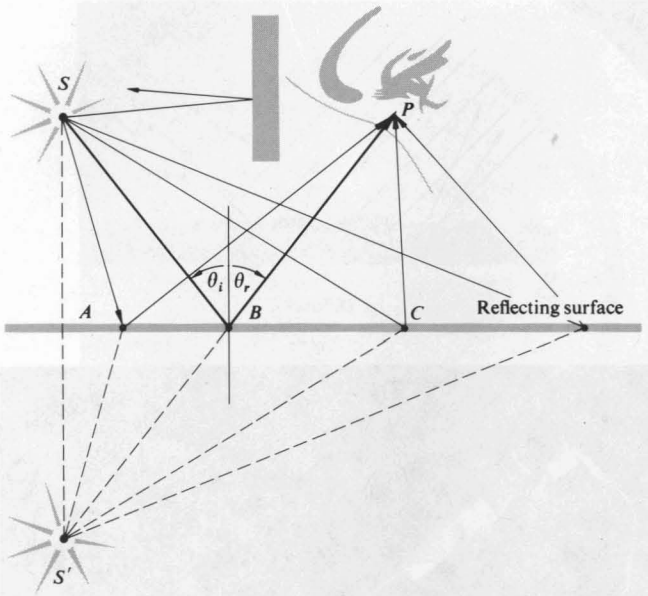


Fig. 4.12 Minimum path from the source S to the observer's eye at P.

what incomplete and a wee bit erroneous at that. For the moment then let us embrace it, but not passionately.

As an example of the application of the principle to the case of refraction refer to Fig. 4.13 where we minimize t, the transit time from S to P, with respect to the variable x. In other words, changing x shifts point O thereby changing the ray from S to P. The smallest transit time will then presumably coincide with the actual path. Hence

$$t = \frac{SO}{v_i} + \frac{OP}{v_t}$$

or

$$t = \frac{(h^2 + x^2)^{1/2}}{v_i} + \frac{[b^2 + (a - x)^2]^{1/2}}{v_t}.$$

To minimize $t(x)$ with respect to variations in x we set $dt/dx = 0$, that is

$$\frac{dt}{dx} = \frac{x}{v_i(h^2 + x^2)^{1/2}} + \frac{-(a - x)}{v_t[b^2 + (a - x)^2]^{1/2}} = 0.$$

Using the diagram this can be rewritten as

$$\frac{\sin \theta_i}{v_i} = \frac{\sin \theta_t}{v_t},$$

which is of course no less than Snell's law (4.4). Thus if a beam of light is to advance from S to P in the least possible time, it must comply with the empirical law of refraction.

Suppose that we have a stratified material composed of m layers each having a different index of refraction as in Fig. 4.14. The transit time from S to P will then be

$$t = \frac{s_1}{v_1} + \frac{s_2}{v_2} + \cdots + \frac{s_m}{v_m}$$

or

$$t = \sum_{i=1}^{m} s_i/v_i,$$

where s_i and v_i are the path length and speed associated with the ith contribution. Thus

$$t = \frac{1}{c}\sum_{i=1}^{m} n_i s_i \tag{4.9}$$

in which the summation is known as the *optical path length* (O.P.L.) traversed by the ray. This is in contrast to the spatial path length $\Sigma_{i=1}^{m} s_i$. Clearly, for an inhomogeneous medium where n is a function of position, the summation must be changed to an integral.

$$(O.P.L.) = \int_S^P n(s)\, ds. \tag{4.10}$$

Inasmuch as $t = (\text{O.P.L.})/c$ we can restate Fermat's principle thus: *light, in going from points S to P, traverses the route*

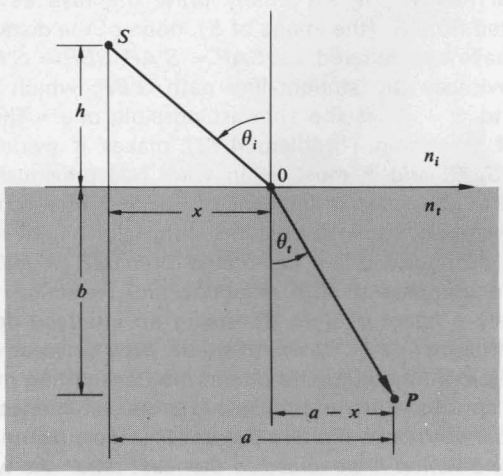


Fig. 4.13 Fermat's principle applied to refraction.

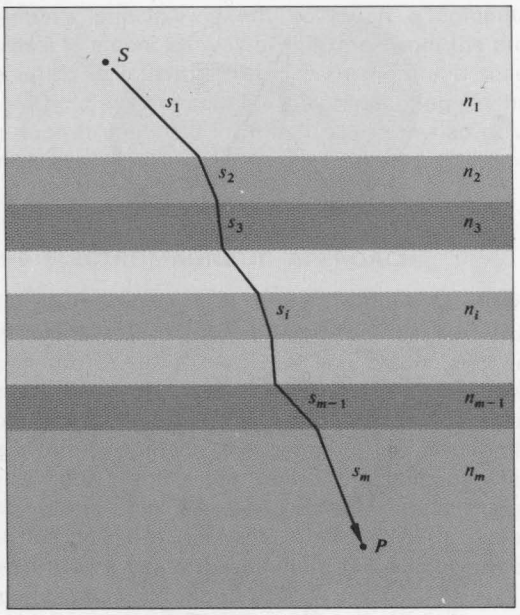


Fig. 4.14 A ray propagating through a layered material.

having the smallest optical path length. Accordingly, when light rays from the sun pass through the inhomogeneous atmosphere of the earth, as shown in Fig. 4.15(a), they bend so as to traverse the lower, denser regions as abruptly as possible, thus minimizing the O.P.L. Ergo, one can still see the sun after it has actually passed below the horizon. In the same way, a road viewed at a glancing angle, as in Fig. 4.15(b), will appear to reflect the environs as if it were covered with a sheet of water. The air near the roadway will be warmer and less dense than that farther above it. Rays will bend upward taking the shortest optical path and in so doing they will seem to an observer as if they had reflected from a mirrored surface. The effect is particularly easy to see on long modern highways. The only requirement is that you look at the road at near glancing incidence because the rays bend very gradually.

The original statement of Fermat's *principle of least time* has some serious failings and is, as we shall see, in need of alteration. To that end, recall that if we have a function, say $f(x)$, we can determine the specific value of the variable x which causes $f(x)$ to have a *stationary* value by setting $df/dx = 0$ and solving for x. By a stationary value we mean one where the slope of $f(x)$ versus x is zero or equivalently where the function has a maximum ⌒, minimum ⌣ or a point of inflection having a horizontal tangent ⌐.

Fermat's principle in its modern form reads: *A light ray in going from point S to point P must traverse an optical path length which is stationary with respect to variations of that path*. In other words, the O.P.L. for the true trajectory will equal, to a first approximation, that O.P.L. of paths immediately adjacent to it.* And so there will be many curves neighboring the actual one which would take very nearly the same time for the light to traverse. This latter point makes it possible to begin to understand how light manages to be so clever in its meanderings. Suppose that we have a beam of light advancing through a homogeneous isotropic medium so that a ray passes from points S to P. Atoms within the material are driven by the incident disturbance and they reradiate in all directions. Quite generally, wavelets originating in the immediate vicinity of a stationary path will arrive at P by routes which differ only slightly and will therefore reinforce each other (see Section 7.1). Wavelets taking other

* The first derivative of the O.P.L. vanishes in its Taylor-series expansion since the path is stationary.

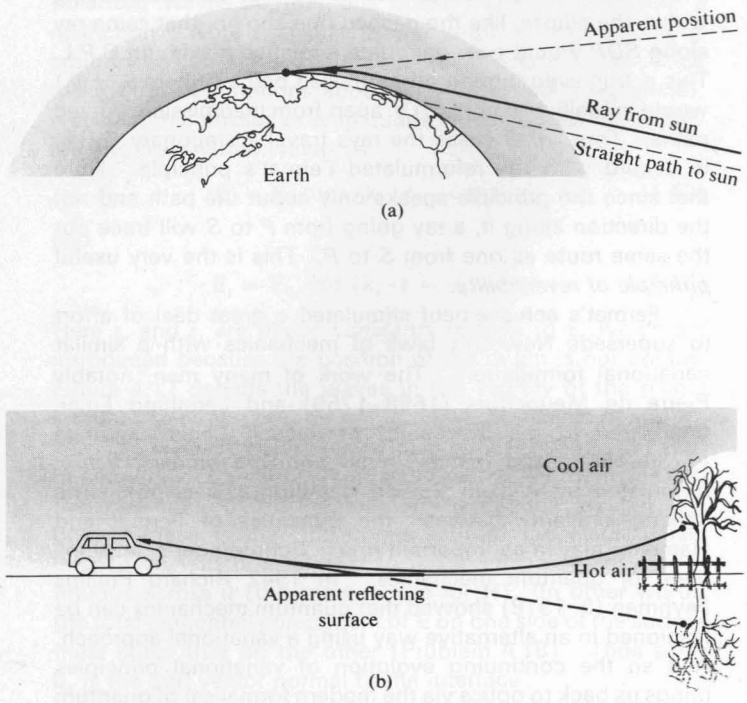
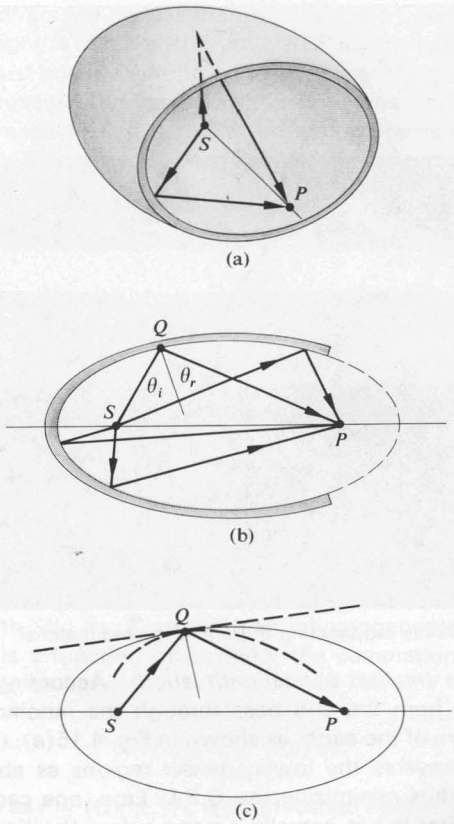


Fig. 4.15 The bending of rays through inhomogeneous media.

paths will arrive at P out of phase and therefore tend to cancel each other. That being the case, energy will effectively propagate along that ray from S to P which satisfies Fermat's principle.

To show that the O.P.L. for a ray need not always be a minimum, examine Fig. 4.16 which depicts a segment of a hollow three-dimensional ellipsoidal mirror. If the source S and the observer P are at the foci of the ellipsoid, then by definition the length SQP will be constant regardless of where on the perimeter Q happens to be. It is also a geometrical property of the ellipse that $\theta_i = \theta_r$ for any location of Q. All optical paths from S to P via a reflection are therefore precisely equal—none is a minimum, and the O.P.L. is clearly stationary with respect to variations. Rays leaving S and striking the mirror will arrive at the focus P. From another viewpoint we can say that radiant energy emitted by S will be scattered by electrons in the mirrored surface such that the wavelets will substantially reinforce each other only at P where they have traveled the same distance and have the same phase. In any case, if there were a plane mirror tangent to the ellipse at Q the exact same path SQP traversed by a ray would then be a relative minimum. At the other extreme, if the mirrored surface conformed to a curve lying within the ellipse, like the dashed one shown, that same ray along SQP would now negotiate a relative maximum O.P.L. This is true even though other unused paths (where $\theta_i \neq \theta_r$) would actually be shorter (i.e. apart from inadmissible curved paths). Thus in all cases the rays travel a stationary O.P.L. in accord with the reformulated Fermat's principle. Note that since the principle speaks only about the path and not the direction along it, a ray going from P to S will trace out the same route as one from S to P. This is the very useful *principle of reversibility*.

Fermat's achievement stimulated a great deal of effort to supersede Newton's laws of mechanics with a similar variational formulation. The work of many men, notably Pierre de Maupertuis (1698–1759) and Leonhard Euler, finally led to the mechanics of Joseph Louis Lagrange (1736–1813) and hence to the *principle of least action* formulated by William Rowan Hamilton (1805–65). The striking similarity between the principles of Fermat and Hamilton played an important role in Schrödinger's development of quantum mechanics. In 1942 Richard Phillips Feynman (b. 1918) showed that quantum mechanics can be fashioned in an alternative way using a variational approach. And so the continuing evolution of variational principles brings us back to optics via the modern formalism of quantum optics (see Chapter 13).



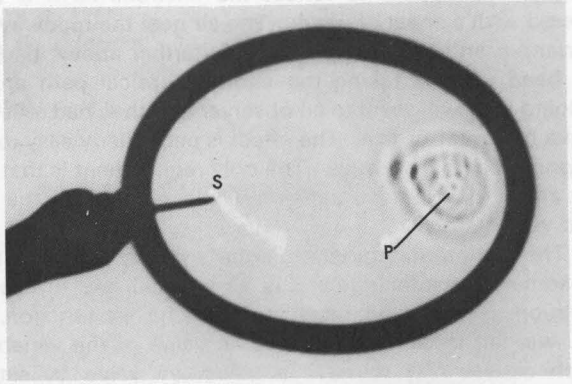


Fig. 4.16 Reflection off an ellipsoidal surface. Observe the reflection of waves using a frying pan filled with water. Even though these are usually circular it's well worth playing with. [Photo courtesy *PSSC College Physics*, D. C. Heath & Co., 1968.]

Fermat's principle is not so much a computational device as it is a concise way of thinking about the propagation of light. It is a statement about the grand scheme of things without any concern for the contributing mechanisms and as such it will yield insights under a myriad of varying circumstances.

4.3 THE ELECTROMAGNETIC APPROACH

Thus far we have been able to deduce the laws of reflection and refraction using three different approaches: *Huygens' principle*, the *theorem of Malus and Dupin* and *Fermat's principle*. Each of these in turn yields a distinctive and valuable point of view of its own. Yet another and even more powerful approach is provided by the electromagnetic theory of light. Unlike the previous techniques which say nothing about the incident, reflected and transmitted radiant flux densities (i.e. I_i, I_r, I_t respectively), the electromagnetic theory treats these within the framework of a far more complete description.

The body of information which forms the subject of optics has accrued over many centuries and as our knowledge of the physical universe becomes more extensive, the concomitant theoretical descriptions must become ever more encompassing. This, quite generally, brings with it an increased complexity. And so, rather than using the formidable mathematical machinery of the quantum theory of light we will most often avail ourselves of the simpler insights of simpler times (e.g. Huygens' and Fermat's principles etc.). Thus even though we are now going to develop still another and more extensive description of reflection and refraction, we will certainly not put aside those earlier methods. In fact, throughout this study we shall use the simplest technique which can yield sufficiently accurate results for our particular purposes.

4.3.1 Waves at an Interface

Suppose that the incident monochromatic light wave is planar so that it has the form

$$\mathbf{E}_i = \mathbf{E}_{0i} \exp\,[i(\mathbf{k}_i \cdot \mathbf{r} - \omega_i t)] \tag{4.11}$$

or more simply

$$\mathbf{E}_i = \mathbf{E}_{0i} \cos\,(\mathbf{k}_i \cdot \mathbf{r} - \omega_i t). \tag{4.12}$$

Let's assume that $\mathbf{E}_{0i}$ is constant in time, i.e. the wave is linearly or plane polarized. We'll find in Chapter 8 that any

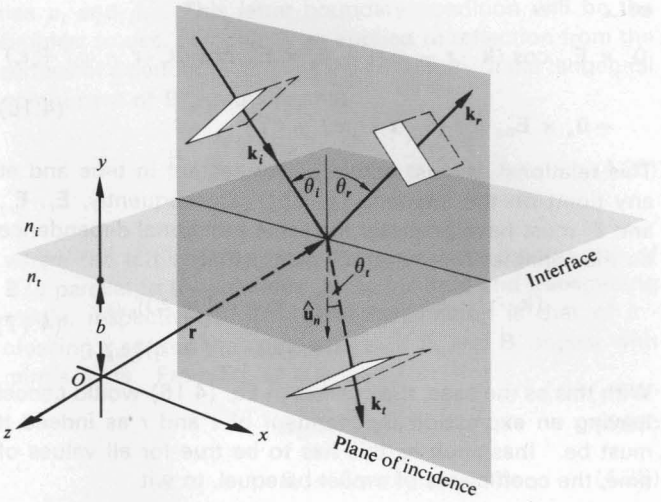


Fig. 4.17 Plane waves incident on the boundary between two dielectric media.

form of light can be represented by two orthogonal linearly polarized waves so that this doesn't actually represent a restriction. Note that just as the origin in time, $t = 0$, is arbitrary so too is the origin O in space where $\mathbf{r} = 0$. Thus making no assumptions about their directions, frequencies, wavelengths, phases or amplitudes, we can write the reflected and transmitted waves as

$$\mathbf{E}_r = \mathbf{E}_{0r} \cos\,(\mathbf{k}_r \cdot \mathbf{r} - \omega_r t + \varepsilon_r) \tag{4.13}$$

and

$$\mathbf{E}_t = \mathbf{E}_{0t} \cos\,(\mathbf{k}_t \cdot \mathbf{r} - \omega_t t + \varepsilon_t). \tag{4.14}$$

Here ε_r and ε_t are *phase constants* relative to $\mathbf{E}_i$ which are introduced because the position of the origin is not unique. Figure 4.17 depicts the waves in the vicinity of the planar interface between two dielectric media of indices n_i and n_t.

The laws of electromagnetic theory (Section 3.1) lead to certain requirements which must be met by the fields and these are referred to as the boundary conditions. Specifically, one of these is that the component of the electric field intensity $\mathbf{E}$ which is tangent to the interface must be continuous across it (the same is true for $\mathbf{H}$). In other words, the total tangential component of $\mathbf{E}$ on one side of the surface must equal that on the other (Problem 4.18). Thus since $\hat{\mathbf{u}}_n$ is the unit vector normal to the interface

$$\hat{\mathbf{u}}_n \times \mathbf{E}_i + \hat{\mathbf{u}}_n \times \mathbf{E}_r = \hat{\mathbf{u}}_n \times \mathbf{E}_t \tag{4.15}$$

or

$$\hat{\mathbf{u}}_n \times \mathbf{E}_{0i} \cos(\mathbf{k}_i \cdot \mathbf{r} - \omega_i t) + \hat{\mathbf{u}}_n \times \mathbf{E}_{0r} \cos(\mathbf{k}_r \cdot \mathbf{r} - \omega_r t + \varepsilon_r) = \hat{\mathbf{u}}_n \times \mathbf{E}_{0t} \cos(\mathbf{k}_t \cdot \mathbf{r} - \omega_t t + \varepsilon_t). \tag{4.16}$$

This relationship must obtain at any instant in time and at any point on the interface ($y = b$). Consequently, $\mathbf{E}_i$, $\mathbf{E}_r$, and $\mathbf{E}_t$ must have precisely the same functional dependence on the variables t and r which means that

$$\begin{aligned}(\mathbf{k}_i \cdot \mathbf{r} - \omega_i t)|_{y=b} &= (\mathbf{k}_r \cdot \mathbf{r} - \omega_r t + \varepsilon_r)|_{y=b} \\ &= (\mathbf{k}_t \cdot \mathbf{r} - \omega_t t + \varepsilon_t)|_{y=b}.\end{aligned} \tag{4.17}$$

With this as the case, the cosines in Eq. (4.16) would cancel leaving an expression independent of t and r as indeed it must be. Inasmuch as this has to be true for all values of time, the coefficients of t must be equal, to wit

$$\omega_i = \omega_r = \omega_t. \tag{4.18}$$

Recall that the electrons within the media are undergoing (linear) forced vibrations at the frequency of the incident wave. Clearly whatever light is scattered has that same frequency. Furthermore

$$(\mathbf{k}_i \cdot \mathbf{r})|_{y=b} = (\mathbf{k}_r \cdot \mathbf{r} + \varepsilon_r)|_{y=b} = (\mathbf{k}_t \cdot \mathbf{r} + \varepsilon_t)|_{y=b}, \tag{4.19}$$

wherein $\mathbf{r}$ terminates on the interface. The values of ε_r and ε_t correspond to a given position of O and thus they allow the relation to be valid regardless of that location. (For example, the origin might be chosen such that $\mathbf{r}$ was perpendicular to $\mathbf{k}_i$ but not $\mathbf{k}_r$ or $\mathbf{k}_t$.) From the first two terms we obtain

$$[(\mathbf{k}_i - \mathbf{k}_r) \cdot \mathbf{r}]_{y=b} = \varepsilon_r. \tag{4.20}$$

Recalling Eq. (2.42), this expression simply says that the endpoint of $\mathbf{r}$ sweeps out a plane (which is of course the interface) perpendicular to the vector $(\mathbf{k}_i - \mathbf{k}_r)$. Phrasing it slightly differently, $(\mathbf{k}_i - \mathbf{k}_r)$ is parallel to $\hat{\mathbf{u}}_n$. Notice however that since the incident and reflected waves are in the same medium $k_i = k_r$. From the fact that $(\mathbf{k}_i - \mathbf{k}_r)$ has no component in the plane of the interface, i.e. $\hat{\mathbf{u}}_n \times (\mathbf{k}_i - \mathbf{k}_r) = 0$, we conclude that

$$k_i \sin\theta_i = k_r \sin\theta_r$$

and hence we have the law of reflection, that is

$$\theta_i = \theta_r.$$

Furthermore since $(\mathbf{k}_i - \mathbf{k}_r)$ is parallel to $\hat{\mathbf{u}}_n$ all three vectors $\mathbf{k}_i$, $\mathbf{k}_r$, and $\hat{\mathbf{u}}_n$ are in the same plane, the plane of incidence. Again, from Eq. (4.19) we obtain

$$[(\mathbf{k}_i - \mathbf{k}_t) \cdot \mathbf{r}]_{y=b} = \varepsilon_t \tag{4.21}$$

and therefore $(\mathbf{k}_i - \mathbf{k}_t)$ is also normal to the interface. Thus $\mathbf{k}_i$, $\mathbf{k}_r$, $\mathbf{k}_t$, and $\hat{\mathbf{u}}_n$ are all coplanar. As before, the tangential components of $\mathbf{k}_i$ and $\mathbf{k}_t$ must be equal and consequently

$$k_i \sin\theta_i = k_t \sin\theta_t. \tag{4.22}$$

But because $\omega_i = \omega_t$ we can multiply both sides by c/ω_i to get

$$n_i \sin\theta_i = n_t \sin\theta_t$$

which is, of course, Snell's law. Finally, observe that had we chosen the origin O to be in the interface it is evident from Eqs. (4.20) and (4.21) that ε_r and ε_t would both have been zero. That arrangement, although not as instructive, is certainly simpler and consequently we'll use it from here on.

4.3.2 Derivation of the Fresnel Equations

We have just found the relationship which exists amongst the phases of $\mathbf{E}_i(\mathbf{r}, t)$, $\mathbf{E}_r(\mathbf{r}, t)$, and $\mathbf{E}_t(\mathbf{r}, t)$ at the boundary. There is still an interdependence shared by the amplitudes $\mathbf{E}_{0i}$, $\mathbf{E}_{0r}$, and $\mathbf{E}_{0t}$ which can now be evaluated. To that end suppose that a plane monochromatic wave is incident on the planar surface separating two isotropic media. Whatever the polarization of the wave we shall resolve its **E**- and **B**-fields into components parallel and perpendicular to the plane of incidence and treat these constituents separately.

Case 1. **E** *Perpendicular to the plane of incidence.* We now assume that **E** is perpendicular to the plane of incidence and that **B** is parallel to it (Fig. 4.18). As you recall $E = vB$ so that

$$\hat{\mathbf{k}} \times \mathbf{E} = v\mathbf{B} \tag{4.23}$$

and of course

$$\hat{\mathbf{k}} \cdot \mathbf{E} = 0, \tag{4.24}$$

i.e. **E**, **B** and the unit propagation vector $\hat{\mathbf{k}}$ form a right-handed system. Again making use of the continuity of the tangential components of the **E**-field, we have at the boundary at any time and any point

$$\mathbf{E}_{0i} + \mathbf{E}_{0r} = \mathbf{E}_{0t} \tag{4.25}$$

where the cosines cancel. We should mention parenthetically that the field vectors as shown really ought to be envisioned at $y = 0$ (i.e. at the surface) from which they have been displaced for the sake of clarity. Note too that while $\mathbf{E}_r$

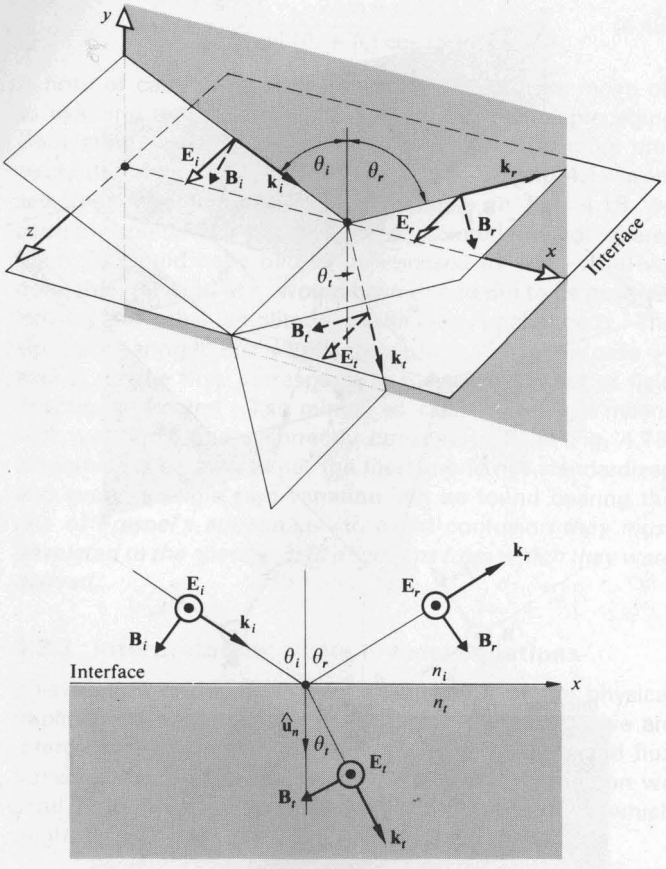


Fig. 4.18 An incoming wave whose **E**-field is normal to the plane of incidence.

and $\mathbf{E}_t$ must be normal to the plane of incidence by symmetry, *we are guessing that they point outward* at the interface when $\mathbf{E}_i$ does. The directions of the **B**-fields then follow from Eq. (4.23).

We will need to invoke another of the boundary conditions in order to get one more equation. The presence of material substances which become electrically polarized by the wave has a definite effect on the field configuration. Thus while the tangential component of **E** is continuous across the boundary, its normal component is not. Instead the normal component of the product $\epsilon\mathbf{E}$ is the same on either side of the interface. Similarly, the normal component of **B** is continuous as is the tangential component of $\mu^{-1}\mathbf{B}$. Here the effect of the two media appears via their permeabilities μ_i and μ_t. This latter boundary condition will be the simplest to use, particularly as applied to reflection from the surface of a conductor.* Thus the continuity of the tangential component of $\mathbf{B}/\mu$ requires that

$$-\frac{\mathbf{B}_i}{\mu_i}\cos\theta_i + \frac{\mathbf{B}_r}{\mu_i}\cos\theta_r = -\frac{\mathbf{B}_t}{\mu_t}\cos\theta_t, \tag{4.26}$$

where the left and right sides are the total magnitudes of $\mathbf{B}/\mu$ parallel to the interface in the incident and transmitting media, respectively. The positive direction is that of increasing x so that the components of $\mathbf{B}_i$ and $\mathbf{B}_t$ appear with minus signs. From Eq. (4.23) we have

$$B_i = E_i/v_i \tag{4.27}$$

$$B_r = E_r/v_r \tag{4.28}$$

and

$$B_t = E_t/v_t. \tag{4.29}$$

Thus since $v_i = v_r$ and $\theta_i = \theta_r$ Eq. (4.26) can be written as

$$\frac{1}{\mu_i v_i}(E_i - E_r)\cos\theta_i = \frac{1}{\mu_t v_t}\mathrm{E}_t\cos\theta_t. \tag{4.30}$$

Making use of Eqs. (4.12), (4.13) and (4.14) and remembering that the cosines therein equal one at $y = 0$ we obtain

$$\frac{n_i}{\mu_i}(E_{0i} - E_{0r})\cos\theta_i = \frac{n_t}{\mu_t}E_{0t}\cos\theta_t. \tag{4.31}$$

Combining this with Eq. (4.25) yields

$$\left(\frac{E_{0r}}{E_{0i}}\right)_\perp = \frac{\dfrac{n_i}{\mu_i}\cos\theta_i - \dfrac{n_t}{\mu_t}\cos\theta_t}{\dfrac{n_i}{\mu_i}\cos\theta_i + \dfrac{n_t}{\mu_t}\cos\theta_t} \tag{4.32}$$

and

$$\left(\frac{E_{0t}}{E_{0i}}\right)_\perp = \frac{2\dfrac{n_i}{\mu_i}\cos\theta_i}{\dfrac{n_i}{\mu_i}\cos\theta_i + \dfrac{n_t}{\mu_t}\cos\theta_t}. \tag{4.33}$$

* In keeping with our intent to use only the **E**- and **B**-fields, at least in the early part of this exposition, we have avoided the usual statements in terms of **H** where

$$\mathbf{H} = \mu^{-1}\mathbf{B} \qquad [A1.14]$$

The $\perp$ subscript serves as a reminder that we are dealing with the case where **E** is perpendicular to the plane of incidence. These two expressions, *which are completely general statements applying to any linear, isotropic, homogeneous media,* are two of what are called the *Fresnel equations*. Quite often one deals with dielectrics for which $\mu_i \approx \mu_t \approx \mu_0$; consequently the most common form of these equations is simply

$$r_\perp \equiv \left(\frac{E_{0r}}{E_{0i}}\right)_\perp = \frac{n_i \cos\theta_i - n_t \cos\theta_t}{n_i \cos\theta_i + n_t \cos\theta_t} \tag{4.34}$$

and

$$t_\perp \equiv \left(\frac{E_{0t}}{E_{0i}}\right)_\perp = \frac{2n_i \cos\theta_i}{n_i \cos\theta_i + n_t \cos\theta_t}. \tag{4.35}$$

Here $r_\perp$ denotes the *amplitude reflection coefficient* while $t_\perp$ is the *amplitude transmission coefficient.*

Case 2. **E** *Parallel to the plane of incidence.* A similar pair of equations can be derived when the incoming **E**-field lies in the plane of incidence as shown in Fig. 4.19. Continuity of the tangential components of **E** on either side of the boundary leads to

$$E_{0i} \cos\theta_i - E_{0r} \cos\theta_r = E_{0t} \cos\theta_t. \tag{4.36}$$

In very much the same way as before, continuity of the tangential components of $\mathbf{B}/\mu$ yields

$$\frac{1}{\mu_i v_i} E_{0i} + \frac{1}{\mu_r v_r} E_{0r} = \frac{1}{\mu_t v_t} E_{0t}. \tag{4.37}$$

Using the fact that $\mu_i = \mu_r$ and $\theta_i = \theta_r$ these formulas can be combined to give us two more of the *Fresnel equations*:

$$r_\parallel \equiv \left(\frac{E_{0r}}{E_{0i}}\right)_\parallel = \frac{\dfrac{n_t}{\mu_t}\cos\theta_i - \dfrac{n_i}{\mu_i}\cos\theta_t}{\dfrac{n_i}{\mu_i}\cos\theta_t + \dfrac{n_t}{\mu_t}\cos\theta_i} \tag{4.38}$$

and

$$t_\parallel \equiv \left(\frac{E_{0t}}{E_{0i}}\right)_\parallel = \frac{2\dfrac{n_i}{\mu_i}\cos\theta_i}{\dfrac{n_i}{\mu_i}\cos\theta_t + \dfrac{n_t}{\mu_t}\cos\theta_i}. \tag{4.39}$$

When both media forming the interface are dielectrics the amplitude coefficients become

$$r_\parallel = \frac{n_t \cos\theta_i - n_i \cos\theta_t}{n_i \cos\theta_t + n_t \cos\theta_i} \tag{4.40}$$

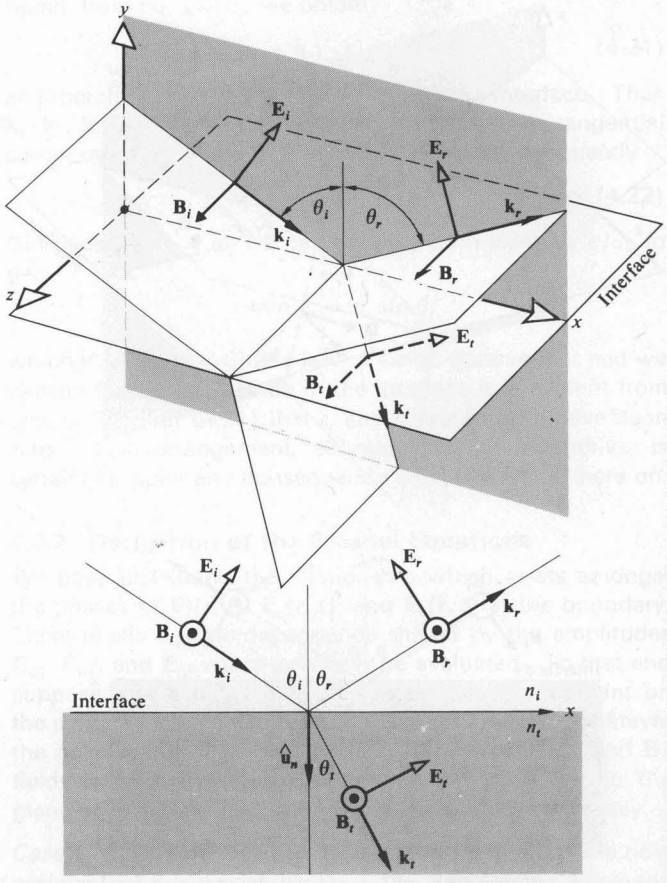


Fig. 4.19 An incoming wave whose **E**-field is in the plane of incidence.

and

$$t_\parallel = \frac{2n_i \cos\theta_i}{n_i \cos\theta_t + n_t \cos\theta_i}. \tag{4.41}$$

One further notational simplification can be made by availing ourselves of Snell's law whereupon the Fresnel equations for dielectric media become (Problem 4.19)

$$r_\perp = -\frac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)} \tag{4.42}$$

$$r_\parallel = +\frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)} \tag{4.43}$$

$$t_\perp = +\frac{2\sin\theta_t \cos\theta_i}{\sin(\theta_i + \theta_t)} \tag{4.44}$$

$$t_{\parallel} = + \frac{2 \sin \theta_t \cos \theta_i}{\sin (\theta_i + \theta_t) \cos (\theta_i - \theta_t)}. \tag{4.45}$$

A note of caution must be introduced before we move on to examine the considerable significance of the preceding calculation. Bear in mind that the directions (or more precisely the phases) of the fields in Figs. 4.18 and 4.19 were selected rather arbitrarily. For example in Fig. 4.18 we certainly could have assumed that $\mathbf{E}_r$ pointed inward, whereupon $\mathbf{B}_r$ would have had to be reversed as well. Had we done that, the sign of $r_{\perp}$ would have turned out to be positive, leaving the other amplitude coefficients unchanged. The signs appearing in Eqs. (4.42) through (4.45), in this case +, except for the first, correspond to the particular set of field directions selected. The minus, as we will see, just means that we didn't guess correctly concerning $\mathbf{E}_r$ in Fig. 4.18. Nonetheless be aware that the literature is not standardized and every possible sign variation can be found bearing the title of *Fresnel's equations*—to avoid confusion *they must be related to the specific field directions from which they were derived.*

4.3.3 Interpretation of the Fresnel equations

This section is devoted to an examination of the physical implications of the Fresnel equations. In particular we are interested in determining the fractional amplitudes and flux densities which are reflected and refracted. In addition we shall be concerned with any possible phase shifts which might be incurred in the process.

i) Amplitude Coefficients

Let's now briefly examine the form of the amplitude coefficients over the entire range of θ_i-values. At nearly normal incidence ($\theta_i \approx 0$) the tangents in Eq. (4.43) are essentially equal to sines in which case

$$[r_{\parallel}]_{\theta_i = 0} = [-r_{\perp}]_{\theta_i = 0} = \left[\frac{\sin (\theta_i - \theta_t)}{\sin (\theta_i + \theta_t)}\right]_{\theta_i = 0}.$$

We will come back to the physical significance of the minus sign presently. After expanding out the sines and using Snell's law this expression becomes

$$[r_{\parallel}]_{\theta_i = 0} = [-r_{\perp}]_{\theta_i = 0} = \left[\frac{n_t \cos \theta_i - n_i \cos \theta_t}{n_t \cos \theta_i + n_i \cos \theta_t}\right]_{\theta_i = 0}. \tag{4.46}$$

which follows as well from Eqs. (4.34) and (4.40). In the limit as θ_i goes to 0, $\cos \theta_i$ and $\cos \theta_t$ both approach one and consequently

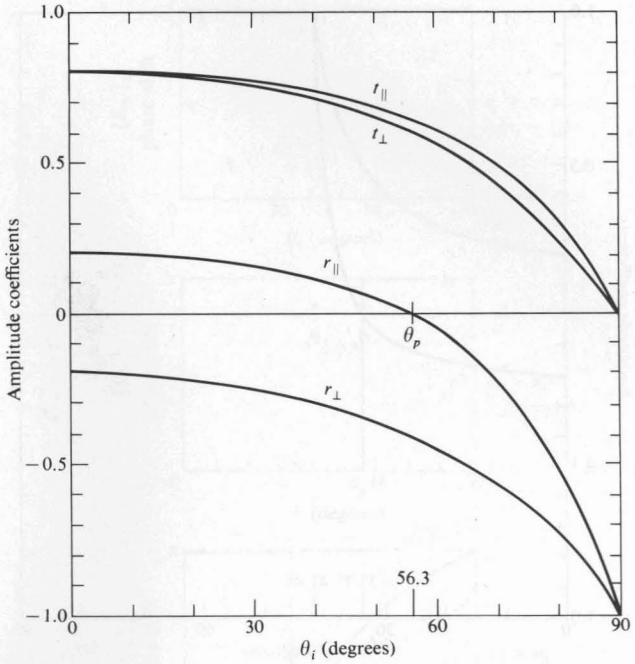


Fig. 4.20 The amplitude coefficients of reflection and transmission as a function of incident angle. These correspond to external reflection $n_t > n_i$ at an air–glass interface ($n_{ti} = 1.5$).

$$[r_{\parallel}]_{\theta_i = 0} = [-r_{\perp}]_{\theta_i = 0} = \frac{n_t - n_i}{n_t + n_i}. \tag{4.47}$$

Thus, for example, at an air ($n_i = 1$) glass ($n_t = 1.5$) interface at near normal incidence the reflection coefficients equal ± 0.2.

When $n_t > n_i$ it follows from Snell's law that $\theta_i > \theta_t$ and so $r_{\perp}$ is negative for all values of θ_i (Fig. 4.20). In contrast, $r_{\parallel}$ starts out positive at $\theta_i = 0$ and decreases gradually until it equals zero when $(\theta_i + \theta_t) = 90°$ since $\tan \pi/2$ is infinite. The particular value of the incident angle for which this occurs is denoted by θ_p and is referred to as the *polarization angle* (see Section 8.6.1). As θ_i increases beyond θ_p, $r_{\parallel}$ becomes ever more negative until it reaches -1.0 at 90°.

At normal incidence Eqs. (4.35) and (4.41) lead, rather straightforwardly, to

$$[t_{\parallel}]_{\theta_i = 0} = [t_{\perp}]_{\theta_i = 0} = \frac{2n_i}{n_i + n_t}. \tag{4.48}$$

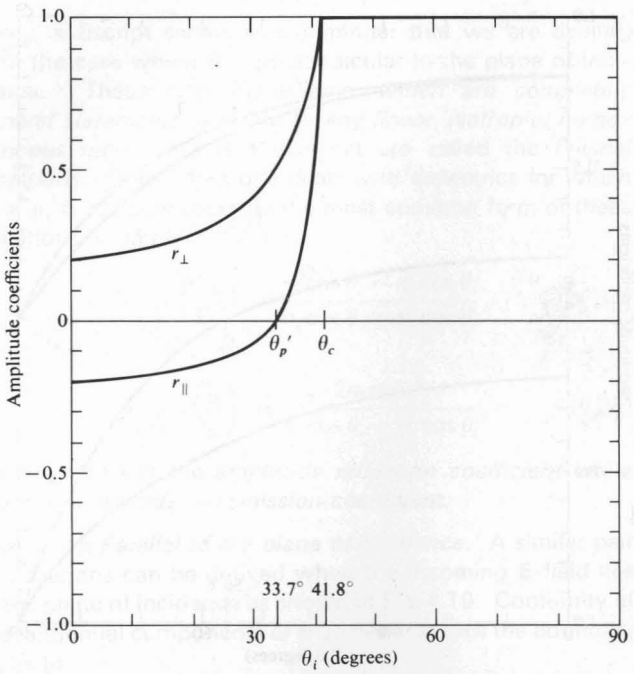


Fig. 4.21 The amplitude coefficients of reflection as a function of incident angle. These correspond to internal reflection $n_t < n_i$ at an air–glass interface ($n_{ti} = 1/1.5$).

It will be shown in Problem 4.21 that the expression

$$t_\perp + (-r_\perp) = 1 \tag{4.49}$$

holds for all θ_i while

$$t_\parallel + r_\parallel = 1 \tag{4.50}$$

is true only at normal incidence.

The foregoing discussion, for the most part, was restricted to the case of *external reflection*, i.e. $n_t > n_i$. The opposite situation of *internal reflection* in which the incident medium is the more dense ($n_i > n_t$) is most certainly of interest as well. In that instance $\theta_t > \theta_i$ and $r_\perp$, as described by Eq. (4.42), will always be positive. As shown in Fig. 4.21, $r_\perp$ increases from its initial value (4.47) at $\theta_i = 0$ reaching plus one at what is called the *critical angle*, θ_c. Specifically, θ_c is the special value of the incident angle for which $\theta_t = \pi/2$. Likewise, $r_\parallel$ starts off negatively (4.47) at $\theta_i = 0$ and thereafter increases going to plus one at $\theta_i = \theta_c$, as is evident from the Fresnel equation (4.40). As before, $r_\parallel$ passes through zero at the *polarization angle* θ_p'. It is left for Problem 4.25 to show that the polarization angles θ_p' and θ_p for internal and external reflection at the interface between the same media are simply the complements of each other. We will return to internal reflection in Section 4.3.4 where it will be shown that $r_\perp$ and $r_\parallel$ are complex quantities for $\theta_i > \theta_c$.

ii) Phase Shifts

It should be evident from Eq. (4.42) that $r_\perp$ is negative regardless of θ_i when $n_t > n_i$. Yet we saw earlier that had we chosen $[\mathbf{E}_r]_\perp$ in Fig. 4.18 to be in the opposite direction, the first Fresnel equation (4.42) would have changed sign causing $r_\perp$ to become a positive quantity. Thus the sign of $r_\perp$ is associated with the relative directions of $[\mathbf{E}_{0i}]_\perp$ and $[\mathbf{E}_{0r}]_\perp$. Bear in mind that a reversal of $[\mathbf{E}_{0r}]_\perp$ is tantamount to introducing a phase shift, $\Delta\varphi_\perp$, of π radians into $[\mathbf{E}_r]_\perp$. Hence at the boundary $[\mathbf{E}_i]_\perp$ and $[\mathbf{E}_r]_\perp$ will be antiparallel and therefore π out of phase with each other as indicated by the negative value of $r_\perp$. When considering components normal to the plane of incidence there is no confusion as to whether two fields are in phase or π radians out of phase; if they are parallel they're in phase; if they're antiparallel they're π out of phase. In summary then, *the component of the electric field normal to the plane of incidence undergoes a phase shift of π radians upon reflection when the incident medium has a lower index than the transmitting medium.* Similarly $t_\perp$ and $t_\parallel$ are always positive and $\Delta\varphi = 0$. *Furthermore, when $n_i > n_t$ no phase shift in the normal component results on reflection, i.e. $\Delta\varphi_\perp = 0$ so long as $\theta_i < \theta_c$.*

Things get a bit less obvious when dealing with $[\mathbf{E}_i]_\parallel$ $[\mathbf{E}_r]_\parallel$ and $[\mathbf{E}_t]_\parallel$. It now becomes necessary to define more explicitly what is meant by *in phase* since the field vectors are coplanar but generally not colinear. The field directions were chosen in Figs. 4.18 and 4.19 such that if you looked down any one of the propagation vectors towards the direction from which the light was coming you would see **E**, **B**, and **k** to have the same relative orientation whether the ray was incident, reflected or transmitted. We can use this as the required condition in order for two **E**-fields to be in phase. Equivalently, but more simply, *two fields in the incident plane are in phase if their y-components are parallel, and out of phase if antiparallel.* Notice that when a pair of **E**-fields are out of phase so too are their associated **B**-fields and vice versa. With this definition we need only look at the vectors normal to the plane of incidence, whether they be **E** or **B**, to determine the relative phase of the accompanying fields in the

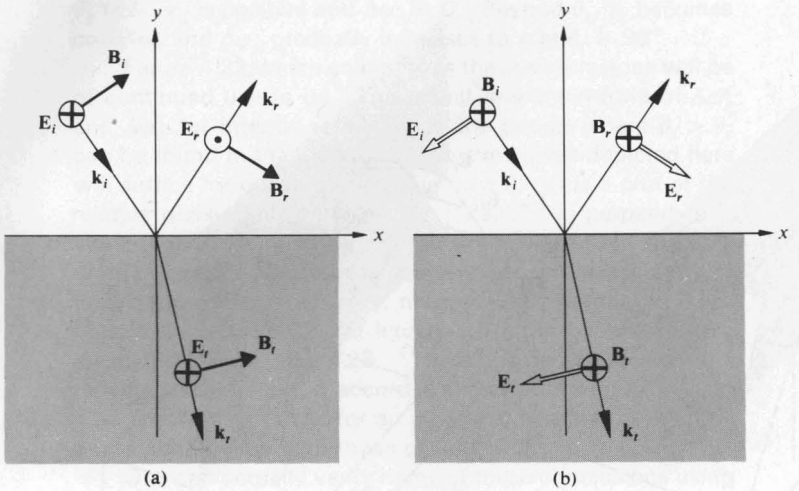


Fig. 4.22 Field orientations and phase shifts.

incident plane. Thus in Fig. 4.22(a) $\mathbf{E}_i$ and $\mathbf{E}_t$ are in phase as are $\mathbf{B}_i$ and $\mathbf{B}_t$ while $\mathbf{E}_i$ and $\mathbf{E}_r$ are out of phase along with $\mathbf{B}_i$ and $\mathbf{B}_r$. Similarly in Fig. 4.22(b) $\mathbf{E}_i$, $\mathbf{E}_r$, and $\mathbf{E}_t$ are in phase as are $\mathbf{B}_i$, $\mathbf{B}_r$, and $\mathbf{B}_t$.

Now, the amplitude reflection coefficient for the parallel component is given by

$$r_{\parallel} = \frac{n_t \cos\theta_i - n_i \cos\theta_t}{n_t \cos\theta_i + n_i \cos\theta_t}$$

which is positive ($\Delta\varphi_{\parallel} = 0$) as long as

$$n_t \cos\theta_i - n_i \cos\theta_t > 0$$

i.e. if

$$\sin\theta_i \cos\theta_i - \cos\theta_t \sin\theta_t > 0$$

or equivalently

$$\sin(\theta_i - \theta_t)\cos(\theta_i + \theta_t) > 0. \tag{4.51}$$

This will be the case for $n_i < n_t$ if

$$(\theta_i + \theta_t) < \pi/2 \tag{4.52}$$

and for $n_i > n_t$ when

$$(\theta_i + \theta_t) > \pi/2. \tag{4.53}$$

And so, when $n_i < n_t$, $[\mathbf{E}_{0r}]_{\parallel}$ and $[\mathbf{E}_{0i}]_{\parallel}$ will be in phase ($\Delta\varphi_{\parallel} = 0$) up until $\theta_i = \theta_p$ and out of phase by π radians thereafter. The transition is not actually discontinuous since $[\mathbf{E}_{0r}]_{\parallel}$ goes to zero at θ_p. In contrast, for internal reflection $r_{\parallel}$ is negative up until θ_p' which means that $\Delta\varphi_{\parallel} = \pi$. From

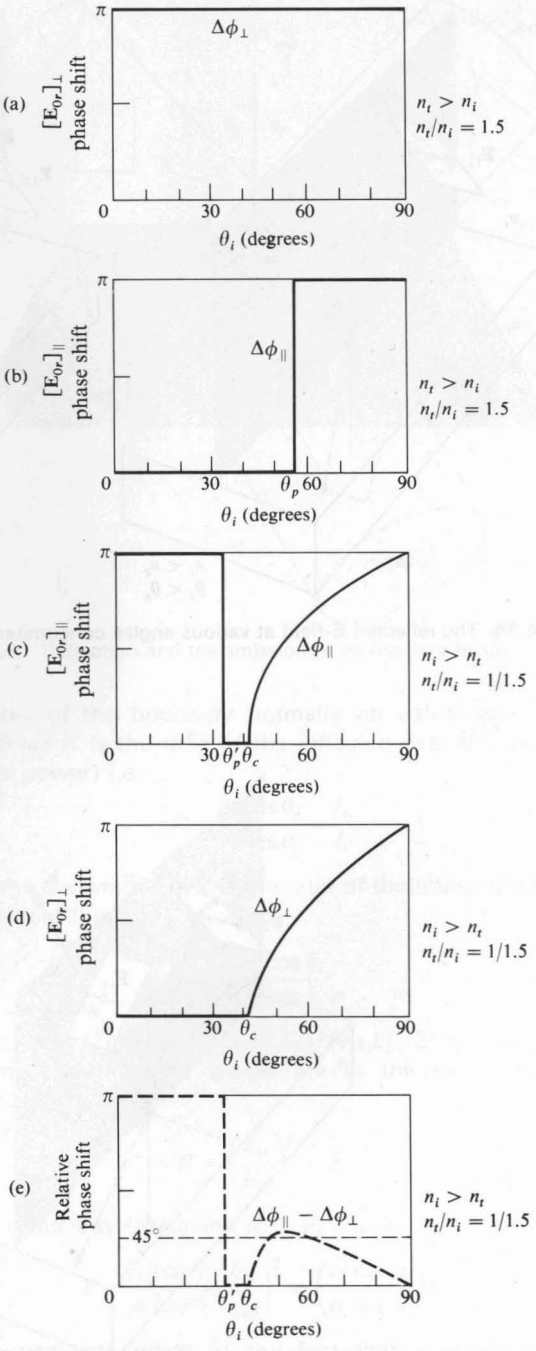


Fig. 4.23 Phase shifts for the parallel and perpendicular components of the **E**-field corresponding to internal and external reflection.

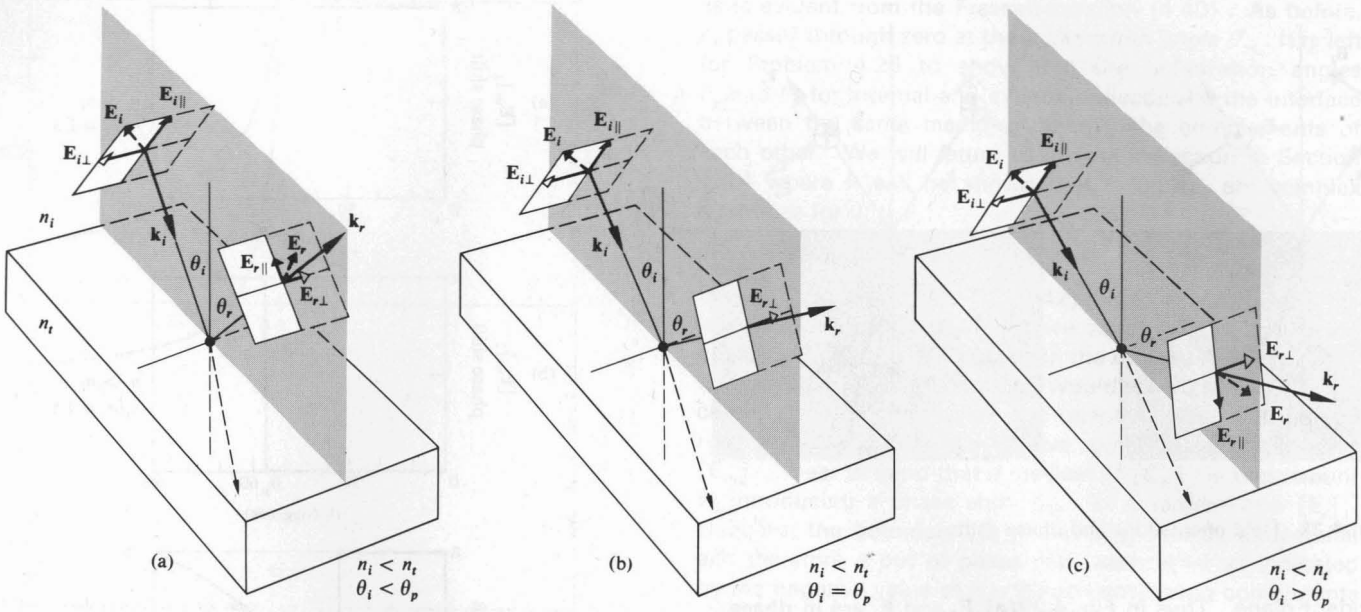


Fig. 4.24 The reflected **E**-field at various angles concomitant with external reflection.

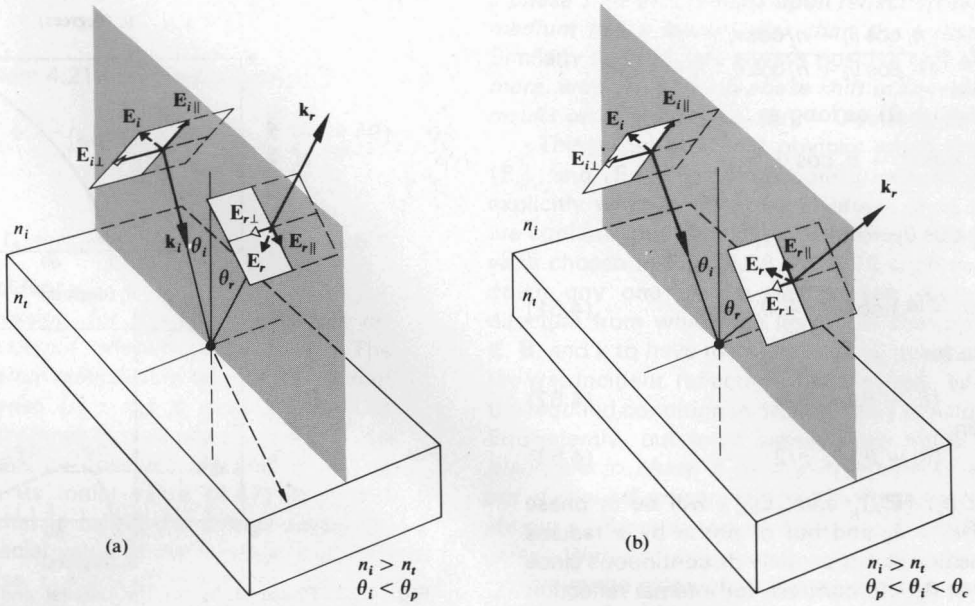


Fig. 4.25 The reflected **E**-field at various angles concomitant with internal reflection.

θ_p' to θ_c, $r_{\parallel}$ is positive and $\Delta\varphi_{\parallel} = 0$. Beyond θ_c, $r_{\parallel}$ becomes complex and $\Delta\varphi_{\parallel}$ gradually increases to π at $\theta_i = 90°$.

Figure 4.23 which summarizes these conclusions will be of continued use to us. The actual functional form of $\Delta\varphi_{\parallel}$ and $\Delta\varphi_{\perp}$ for internal reflection in the region where $\theta_i > \theta_c$ can be found in the literature,* but, the curves depicted here will suffice for our purposes. Figure 4.23(e) is a plot of the relative phase shift between the parallel and perpendicular components, i.e. $\Delta\varphi_{\parallel} - \Delta\varphi_{\perp}$. It is included here because we anticipate its usefulness later on, e.g. when we consider polarization effects. Finally, many of the essential features of this discussion can be illustrated rather unambiguously as in Figs. 4.24 and 4.25. There the amplitudes of the reflected vectors are in accord with those of Figs. 4.20 and 4.21 (recall that this is for an air–glass interface) while the phase shifts agree with those of Fig. 4.23.

One can actually verify many of these conclusions using the simplest experimental facilities, namely two linear polarizers, a piece of glass and a small source such as a flashlight or "high intensity" lamp. By placing one polarizer in front of the source (at 45° to the plane of incidence) you can easily duplicate the conditions of Fig. 4.24. For example, when $\theta_i = \theta_p$ [Fig. 4.24(b)] no light will pass through the second polarizer if its transmission axis is parallel to the plane of incidence. In comparison, at near glancing incidence the reflected beam will vanish when the axes of the two polarizers are almost normal to each other.

iii) Reflectance and Transmittance

Recall that the power per unit area crossing a surface in vacuum whose normal is parallel to **S**, the Poynting vector, is given by

$$\mathbf{S} = c^2\epsilon_0 \mathbf{E} \times \mathbf{B}. \qquad [3.48]$$

Furthermore, the radiant flux density (W/m^2) or irradiance is then

$$I = \langle S \rangle = \frac{c\epsilon_0}{2} E_0^2. \qquad [3.52]$$

This is the average energy per unit time crossing a unit area normal to **S** (in isotropic media **S** is parallel to **k**). In the case at hand (Fig. 4.26), let I_i, I_r, and I_t be the incident, reflected and transmitted flux densities respectively. Accordingly, the portion of the energy incident *normally on a unit area of the boundary* per second is $I_i \cos\theta_i$. Similarly, $I_r \cos\theta_r$ and $I_t \cos\theta_t$ are the energies per second leaving a

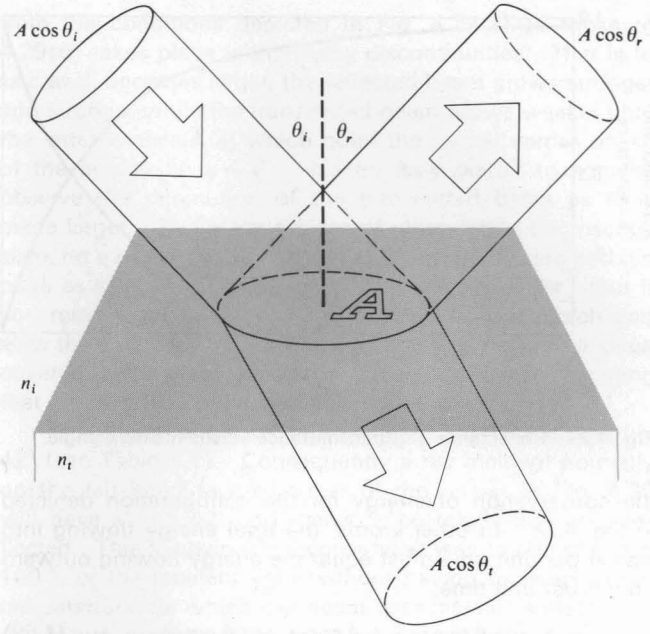


Fig. 4.26 Reflection and transmission of an incident beam.

unit area of the boundary normally on either side. The *reflectance* R is the ratio of the reflected over the incident flux (or power) i.e.

$$R \equiv \frac{I_r \cos\theta_r}{I_i \cos\theta_i} = \frac{I_r}{I_i}, \qquad (4.54)$$

while the *transmittance* T is the ratio of the transmitted over the incident flux and is given by

$$T \equiv \frac{I_t \cos\theta_t}{I_i \cos\theta_i}. \qquad (4.55)$$

The quotient I_r/I_i equals $(v_r\epsilon_r E_{0r}^2/2)/(v_i\epsilon_i E_{0i}^2/2)$ and since the incident and reflected waves are in the same medium $v_r = v_i$, $\epsilon_r = \epsilon_i$, and

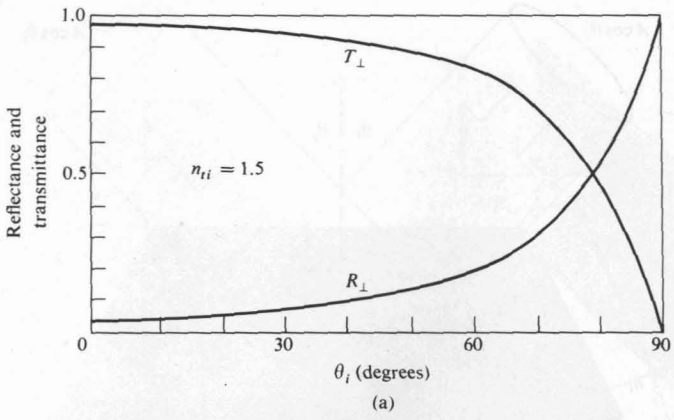
$$R = \left(\frac{E_{0r}}{E_{0i}}\right)^2 = r^2. \qquad (4.56)$$

In the same way (assuming $\mu_i = \mu_t = \mu_0$)

$$T = \frac{n_t \cos\theta_t}{n_i \cos\theta_i}\left(\frac{E_{0t}}{E_{0i}}\right)^2 = \left(\frac{n_t \cos\theta_t}{n_i \cos\theta_i}\right) t^2, \qquad (4.57)$$

where use was made of the fact that $\mu_0\epsilon_t = 1/v_t^2$ and $\mu_0 v_t \epsilon_t = n_t/c$. Let's now write an expression representing

* Born and Wolf, *Principles of Optics*, p. 49.



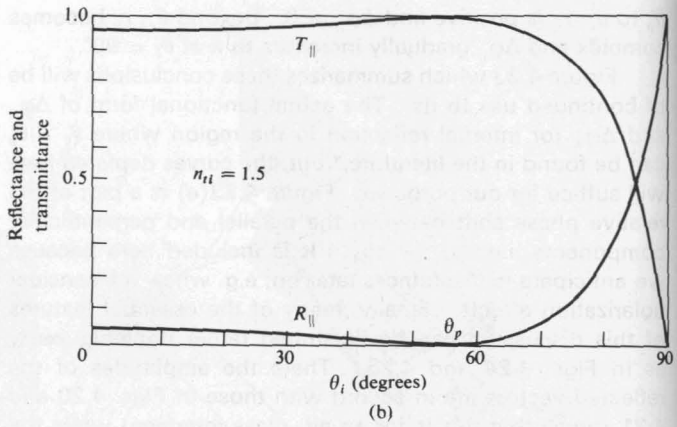


Fig. 4.27 Reflectance and transmittance versus incident angle.

the conservation of energy for the configuration depicted in Fig. 4.26. In other words, the total energy flowing into area A per unit time must equal the energy flowing outward from it per unit time;

$$I_i A \cos\theta_i = I_r A \cos\theta_r + I_t A \cos\theta_t. \tag{4.58}$$

It is evident here as in Fig. 4.8 that the cross sectional area of the transmitted beam, $A\cos\theta_t$ is larger than that of the incident or reflected beams (which are equal). Multiplying both sides by c this expression becomes

$$n_i E_{0i}^2 \cos\theta_i = n_i E_{0r}^2 \cos\theta_i + n_t E_{0t}^2 \cos\theta_t$$

or

$$1 = \left(\frac{E_{0r}}{E_{0i}}\right)^2 + \left(\frac{n_t \cos\theta_t}{n_i \cos\theta_i}\right)\left(\frac{E_{0t}}{E_{0i}}\right)^2. \tag{4.59}$$

But this is simply

$$R + T = 1 \tag{4.60}$$

where there was no absorption. It is convenient to use the component forms, that is

$$R_\perp = r_\perp^2 \tag{4.61}$$

$$R_\parallel = r_\parallel^2 \tag{4.62}$$

$$T_\perp = \left(\frac{n_t \cos\theta_t}{n_i \cos\theta_i}\right) t_\perp^2 \tag{4.63}$$

and

$$T_\parallel = \left(\frac{n_t \cos\theta_t}{n_i \cos\theta_i}\right) t_\parallel^2 \tag{4.64}$$

which are illustrated in Fig. 4.27. Furthermore, it can be shown (Problem 4.30) that

$$R_\parallel + T_\parallel = 1 \tag{4.65}$$

and

$$R_\perp + T_\perp = 1. \tag{4.66}$$

One interesting feature of these curves is easily verified at least qualitatively, and that is that both $R_\parallel$ and $R_\perp$ approach one as $\theta_i \to 90°$. This implies that almost any fairly smooth dielectric surface will actually become mirror-like at glancing incidence. Try looking at a light source using this page as a boundary surface where $\theta_i \approx 90°$. You should be able to see a rather clear image of the source reflected in the paper.

When $\theta_i = 0$ the incident plane becomes undefined and any distinction between the parallel and perpendicular components of R and T vanishes. In this case Eqs. (4.61) through (4.64) along with (4.47) and (4.48) lead to

$$R = R_\parallel = R_\perp = \left(\frac{n_t - n_i}{n_t + n_i}\right)^2 \tag{4.67}$$

and

$$T = T_\parallel = T_\perp = \frac{4 n_t n_i}{(n_t + n_i)^2}. \tag{4.68}$$

Thus 4% of the light incident normally on an air–glass interface will be reflected back whether internally, $n_i > n_t$, or externally, $n_i < n_t$ (Problem 4.31). This will obviously become of great concern to anyone who is working with a complicated lens system which might have ten or twenty such air–glass boundaries. Indeed, if you look perpendicularly into a stack of about fifty microscope slides, (cover-

glass slides are much thinner and easier to handle in large quantities) most of the light will be reflected. The stack will appear very much like a mirror (Fig. 4.28).

4.3.4 Total Internal Reflection

In the previous section it was evident that something rather interesting was happening in the case of internal reflection ($n_i > n_t$) when θ_i was equal to or greater than θ_c, the so-called *critical angle*. Let's now return to that situation for a somewhat closer look at things. Suppose that we have a source imbedded in an optically dense medium and we allow θ_i to gradually increase as indicated in Fig. 4.29. We know from the preceding section (Fig. 4.21) that $r_{\parallel}$ and $r_{\perp}$ increase with increasing θ_i and therefore $t_{\parallel}$ and $t_{\perp}$ both decrease. Moreover $\theta_t > \theta_i$ since

$$\sin \theta_i = \frac{n_t}{n_i} \sin \theta_t$$

and $n_i > n_t$ in which case $n_{ti} < 1$. Thus as θ_i becomes larger the transmitted ray gradually approaches tangency with the boundary and as it does so more and more of the available energy appears in the reflected beam. Finally when $\theta_t = 90°$, $\sin \theta_t = 1$ and

$$\sin \theta_c = n_{ti}. \tag{4.69}$$

As was pointed out earlier, *the critical angle is that special value of θ_i for which $\theta_t = 90°$*. For incident angles greater than or equal to θ_c all of the incoming energy is reflected back into the incident medium in the process known as *total internal reflection*. It should be stressed that the transition from the conditions depicted in Fig. 4.29(a) to those of 4.29(d) takes place without any discontinuities. That is to say, as θ_i becomes larger, the reflected beam grows stronger and stronger while the transmitted beam grows weaker until the latter vanishes, at which point the former carries off all of the energy at $\theta_r = \theta_c$. It's an easy matter to actually observe the diminution of the transmitted beam as θ_i is made larger. Just place a piece of glass, e.g. a microscope slide, on a printed page. At $\theta_i \approx 0$, θ_t is roughly zero and the page as seen through the glass is bright and clear. But if you move your head allowing θ_t (the angle at which you view the interface) to increase, the region of the printed page covered by the glass will appear darker and darker indicating that T has indeed been markedly reduced.

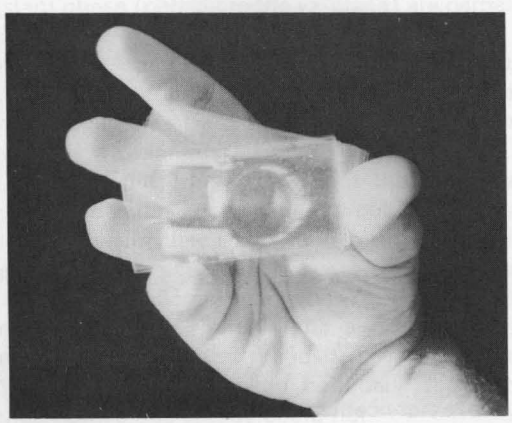


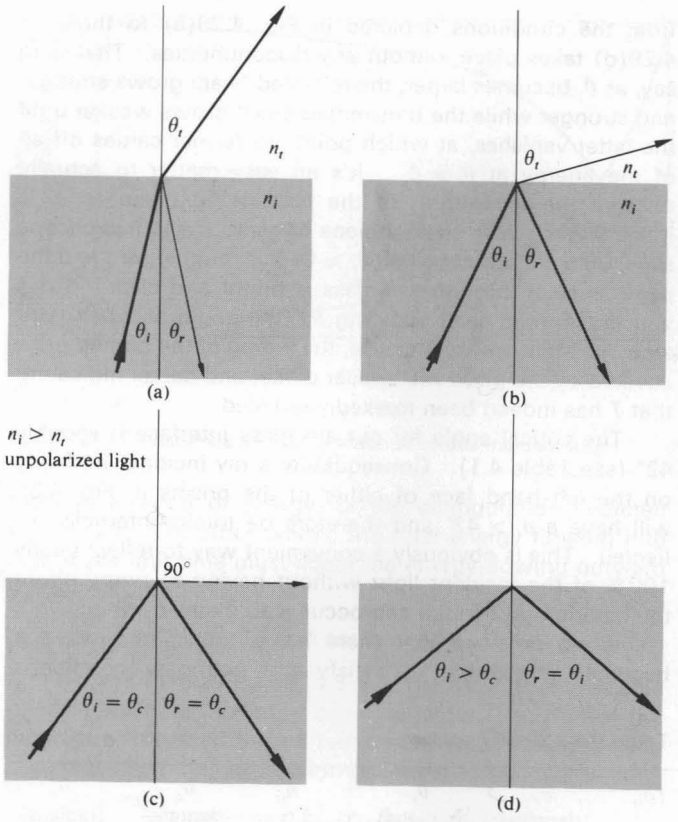
Fig. 4.28 Near normal reflection off a stack of microscope slides. You can see the image of the camera which took the picture. [Photo by E.H.]

The critical angle for our air–glass interface is roughly 42° (see Table 4.1). Consequently a ray incident normally on the left-hand face of either of the prisms in Fig. 4.30 will have a $\theta_i > 42°$ and therefore be totally internally reflected. This is obviously a convenient way to reflect nearly 100% of the incident light without having to worry about the deterioration which can occur with metallic surfaces.

If we presume that there is no transmitted wave it becomes impossible to satisfy the boundary conditions

Table 4.1 Critical angles.

n_{it}	θ_c (degrees)	θ_c (radians)	n_{it}	θ_c (degrees)	θ_c (radians)
1.30	50.2849	0.8776	1.50	41.8103	0.7297
1.31	49.7612	0.8685	1.51	41.4718	0.7238
1.32	49.2509	0.8596	1.52	41.1395	0.7180
1.33	48.7535	0.8509	1.53	40.8132	0.7123
1.34	48.2682	0.8424	1.54	40.4927	0.7067
1.35	47.7946	0.8342	1.55	40.1778	0.7012
1.36	47.3321	0.8261	1.56	39.8683	0.6958
1.37	46.8803	0.8182	1.57	39.5642	0.6905
1.38	46.4387	0.8105	1.58	39.2652	0.6853
1.39	46.0070	0.8030	1.59	38.9713	0.6802
1.40	45.5847	0.7956	1.60	38.6822	0.6751
1.41	45.1715	0.7884	1.61	38.3978	0.6702
1.42	44.7670	0.7813	1.62	38.1181	0.6653
1.43	44.3709	0.7744	1.63	37.8428	0.6605
1.44	43.9830	0.7676	1.64	37.5719	0.6558
1.45	43.6028	0.7610	1.65	37.3052	0.6511
1.46	43.2302	0.7545	1.66	37.0427	0.6465
1.47	42.8649	0.7481	1.67	36.7842	0.6420
1.48	42.5066	0.7419	1.68	36.5296	0.6376
1.49	42.1552	0.7357	1.69	36.2789	0.6332



using only the incident and reflected waves—things are not at all as simple as they might seem. Furthermore we can reformulate Eqs. (4.34) and (4.40) (Problem 4.34) such that

$$r_\perp = \frac{\cos\theta_i - (n_{ti}^2 - \sin^2\theta_i)^{\frac{1}{2}}}{\cos\theta_i + (n_{ti}^2 - \sin^2\theta_i)^{\frac{1}{2}}} \tag{4.70}$$

and

$$r_{\|} = \frac{n_{ti}^2\cos\theta_i - (n_{ti}^2 - \sin^2\theta_i)^{\frac{1}{2}}}{n_{ti}^2\cos\theta_i + (n_{ti}^2 - \sin^2\theta_i)^{\frac{1}{2}}}. \tag{4.71}$$

Clearly then since $\sin\theta_c = n_{ti}$ when $\theta_i > \theta_c$, $\sin\theta_i > n_{ti}$ and both $r_\perp$ and $r_{\|}$ become complex quantities. Despite this (Problem 4.35) $r_\perp r_\perp^* = r_{\|} r_{\|}^* = 1$ and $R = 1$ which means that $I_r = I_i$ and $I_t = 0$. Thus, although there must be a transmitted wave it cannot, on the average, carry energy across the boundary. We shall not perform the complete and rather lengthy computation needed to derive expressions for all of the reflected and transmitted fields but we can get an

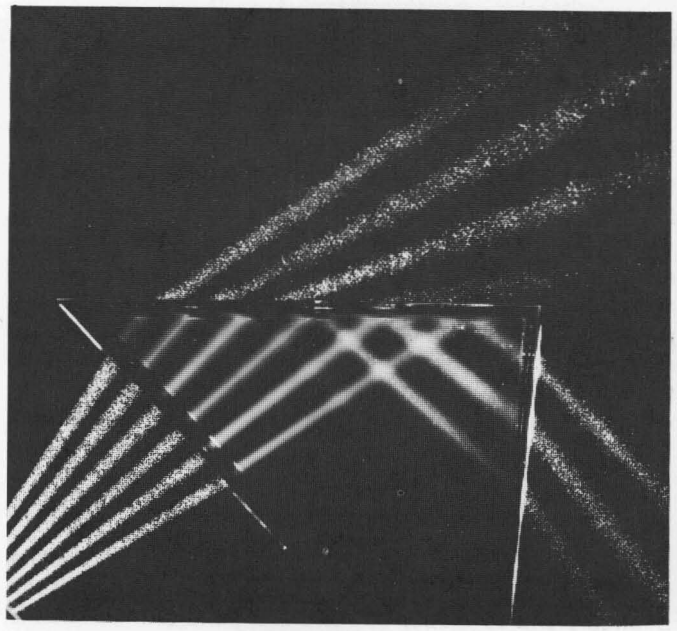


Fig. 4.29 Internal reflection and the critical angle. [Photo courtesy of Educational Service, Inc.]

appreciation of what's happening in the following way. The wave function for the transmitted electric field is

$$\mathbf{E}_t = \mathbf{E}_{0t}\exp i(\mathbf{k}_t\cdot\mathbf{r} - \omega t)$$

where

$$\mathbf{k}_t\cdot\mathbf{r} = k_{tx}x + k_{ty}y,$$

there being no z-component of $\mathbf{k}$. But

$$k_{tx} = k_t\sin\theta_t$$

and

$$k_{ty} = k_t\cos\theta_t$$

as can be seen from Fig. 4.31. Once again using Snell's law

$$k_t\cos\theta_t = \pm k_t\left(1 - \frac{\sin^2\theta_i}{n_{ti}^2}\right)^{\frac{1}{2}} \tag{4.72}$$

or since we are concerned with the case where $\sin\theta_i > n_{ti}$,

$$k_{ty} = \pm ik_t\left(\frac{\sin^2\theta_i}{n_{ti}^2} - 1\right)^{\frac{1}{2}} \equiv \pm i\beta$$

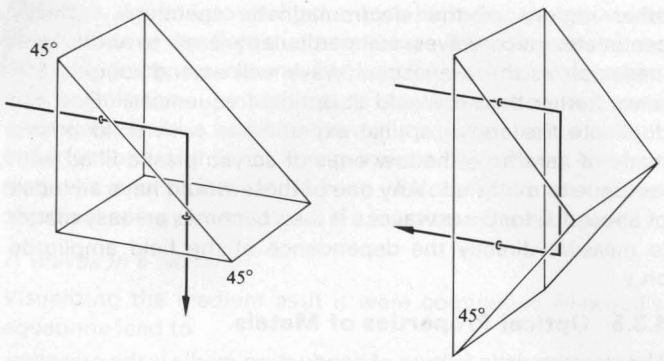


Fig. 4.30 Total internal reflection.

while

$$k_{tx} = \frac{k_t}{n_{ti}} \sin \theta_i.$$

Hence

$$\mathbf{E}_t = \mathbf{E}_{0t} e^{\mp \beta y} e^{i(k_t x \sin \theta_i / n_{ti} - \omega t)}. \tag{4.73}$$

Neglecting the positive exponential which is physically untenable we have a wave whose amplitude drops off exponentially as it penetrates the less dense medium. The disturbance advances in the x-direction as a so-called *surface* or *evanescent wave*. Notice that the wavefronts or surfaces of constant phase (parallel to the yz-plane) are perpendicular to the surfaces of constant amplitude (parallel to the xz-plane) and as such the wave is *inhomogeneous* (see Section 2.5). Its amplitude decays rapidly in the y-direction, becoming negligible at a distance into the second medium of onlv a few wavelengths.

If you are still concerned about the conservation of energy, a more extensive treatment would have shown that energy actually circulates back and forth across the interface resulting on the average in a zero net flow through the boundary into the second medium. Yet one puzzling point remains inasmuch as there is still a bit of energy to be accounted for, namely that associated with the evanescent wave which moves along the boundary in the plane of incidence. Since this energy could not have penetrated into the less dense medium under the present circumstances (so long as $\theta_i \geq \theta_c$) we must look elsewhere for its source. Under actual experimental conditions the incident beam would have a finite cross-section and therefore would obviously differ from a true plane wave. This deviation gives rise (via diffraction) to a slight transmission of energy across the interface which is manifest in the evanescent wave.

Incidently, it is clear from Fig. 4.23(c) and (d) that the incident and reflected waves (except at $\theta_i = 90°$) do not differ in phase by π and cannot therefore cancel each other. It follows from the continuity of the tangential component of **E** that there must therefore be an oscillatory field in the less dense medium with a component parallel to the interface having a frequency ω (i.e. the evanescent wave).

The exponential decay of the surface or *boundary wave*, as it is also sometimes called, has quite recently been confirmed experimentally at optical frequencies.*

Imagine that a beam of light traveling within a block of glass is internally reflected at a boundary. Presumably if we pressed another piece of glass against the first the air–glass interface could be made to vanish and the beam would then propagate onward undisturbed. Furthermore, we might expect this transition from total to no reflection to occur gradually as the air film thins out. In much the same way, if one holds a drinking glass or a prism you can see the ridges of your fingerprints in a region which, because of total internal reflection, is otherwise mirror like. In more general terms, if the evanescent wave extends with appreciable amplitude across the rare medium into a nearby region occupied by a higher-index material, energy may now flow through the

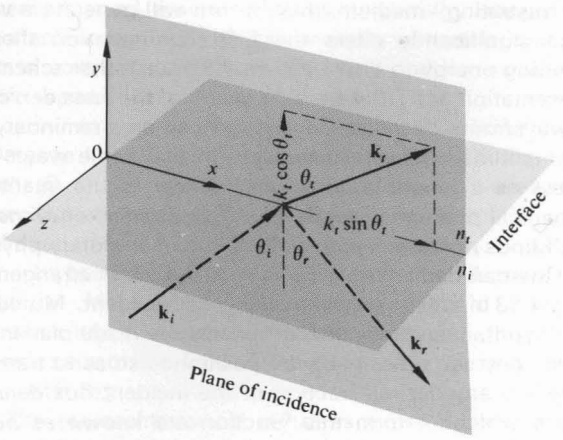


Fig. 4.31 Propagation vectors for internal reflection.

* Take a look at the fascinating article "Monomolecular Layers and Light" by K. H. Drexhage, *Sci. Am.* **222**, 108 (1970).

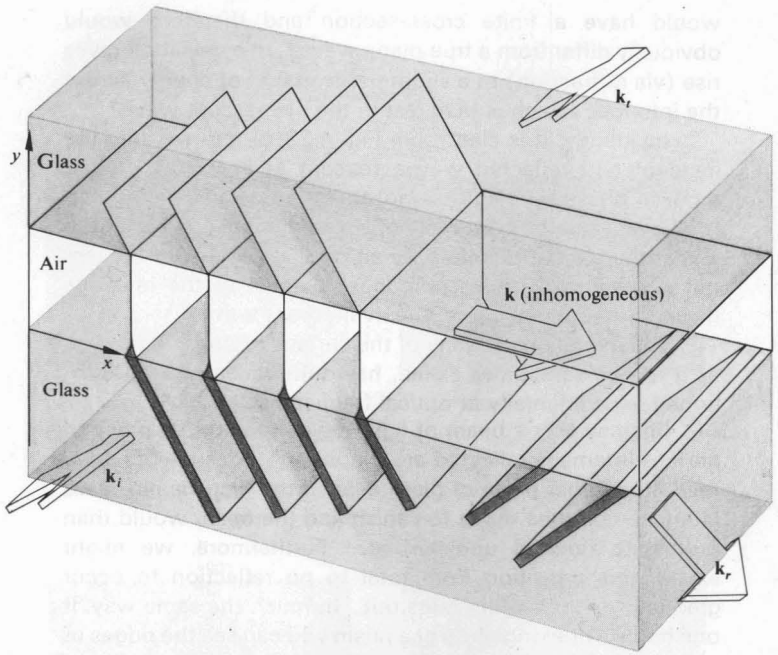


Fig. 4.32 Frustrated total internal reflection.

gap in what is known as *frustrated total internal reflection* (FTIR). In other words, if the evanescent wave, having traversed the gap, is still strong enough to drive electrons in the "frustrating" medium, they in turn will generate a wave which significantly alters the field configuration thereby permitting energy to flow. Figure 4.32 is a rather schematic representation of FTIR where the width of the lines depicting the wavefronts decreases across the gap as a reminder that the amplitude of the field behaves in that same way. The process as a whole is remarkably similar to the quantum-mechanical phenomenon of *barrier penetration* or *tunneling* which finds numerous applications in contemporary physics.

One can demonstrate FTIR with the prism arrangement of Fig. 4.33 in a manner which is fairly self-evident. Moreover, if the hypotenuse faces of both prisms are made planar and parallel they can presumably be positioned so as to transmit and reflect any desired fraction of the incident flux density. Devices which perform this function are known as *beam splitters*. *A beam splitter cube* can be made rather conveniently by using a thin low index transparent film as a precision spacer. Low-loss reflectors whose transmittance can be controlled by frustrating internal reflection are of considerable practical interest. FTIR can also be observed in other regions of the electromagnetic spectrum. Three-centimeter microwaves are particularly easy to work with inasmuch as the evanescent wave will extend roughly 10^5 times farther than it would at optical frequencies. One can duplicate the above optical experiments with solid prisms made of paraffin or hollow ones of acrylic plastic filled with kerosene or motor oil. Any one of these would have an index of about 1.5 for 3 cm waves. It then becomes an easy matter to measure directly the dependence of the field amplitude on y.

4.3.5 Optical Properties of Metals

The characteristic feature of conducting media is the presence of a number of free electric charges (free in the sense of being unbound i.e. able to circulate around within the material). For metals these charges are of course electrons and their motion constitutes a current. The current per unit area resulting from the application of a field **E** is related by way of Eq. (A1.15) to the conductivity of the medium σ. For a dielectric there are no free or conduction electrons and $\sigma = 0$ while for actual metals σ is nonzero and finite. In contrast, an idealized "perfect" conductor would have an infinite conductivity. This is equivalent to saying that the

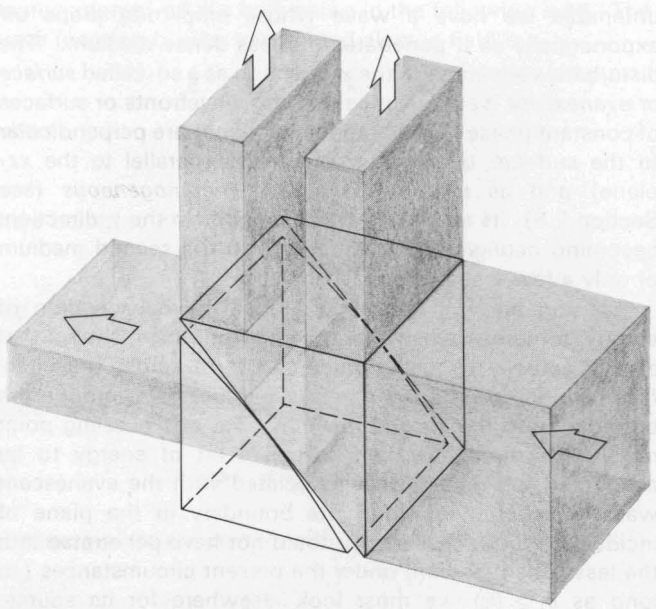


Fig. 4.33 A beam splitter utilizing FTIR.

electrons, driven into oscillation by a harmonic wave, would simply follow the field's alternations. There would be no restoring force, no natural frequencies and no absorption, only reemission. In real metals the conduction electrons undergo collisions with the thermally agitated lattice or with imperfections and in so doing irreversibly convert electromagnetic energy into Joule heat. Evidently the absorption of radiant energy by a material is a function of its conductivity.

i) Waves in a Metal

Visualizing the medium as if it were continuous Maxwell's equations lead to

$$\frac{\partial^2 \mathbf{E}}{\partial x^2} + \frac{\partial^2 \mathbf{E}}{\partial y^2} + \frac{\partial^2 \mathbf{E}}{\partial z^2} = \mu\epsilon\frac{\partial^2 \mathbf{E}}{\partial t^2} + \mu\sigma\frac{\partial \mathbf{E}}{\partial t} \tag{4.74}$$

which is Eq. (A1.21) in Cartesian coordinates. The last term, $\mu\sigma\,\partial\mathbf{E}/\partial t$, is a first-order time derivative as was the damping force in the oscillator model of Section 3.3.1. The time rate of change of **E** generates a voltage, currents circulate and since the material is resistive, light is converted to heat—ergo absorption. It can be shown that this expression reduces to the unattenuated wave equation if the permittivity is reformulated as a complex quantity. This in turn leads to a complex index of refraction which, as we saw earlier (Section 3.3.1), is tantamount to absorption. We then need only substitute the complex index

$$n_c = n_R - in_I \tag{4.75}$$

(where the real and imaginary indices n_R and n_I are both real numbers) into the corresponding solution for a nonconducting medium. Alternatively, one may utilize the wave equation and appropriate boundary conditions to yield a specific solution. In either event, we can find a simple sinusoidal plane-wave solution applicable within the conductor. Such a wave propagating in the y-direction is ordinarily written as

$$\mathbf{E} = \mathbf{E}_0 \cos(\omega t - ky)$$

or as a function of n

$$\mathbf{E} = \mathbf{E}_0 \cos \omega(t - ny/c)$$

but here the refractive index must be taken as complex. Accordingly, writing the wave as an exponential and using Eq. (4.75) it becomes

$$\mathbf{E} = \mathbf{E}_0 e^{(-\omega n_I y/c)} e^{i\omega(t - n_R y/c)} \tag{4.76}$$

or

$$\mathbf{E} = \mathbf{E}_0 e^{-\omega n_I y/c} \cos \omega(t - n_R y/c). \tag{4.77}$$

The disturbance advances in the y-direction with a speed c/n_R precisely as if n_R were the more usual index of refraction. As the wave progresses into the conductor, its amplitude, $\mathbf{E}_0 \exp(-\omega n_I y/c)$, is exponentially attenuated. Inasmuch as irradiance is proportional to the square of the amplitude, we have

$$I(y) = I_0 e^{-\alpha y}, \tag{4.78}$$

where $I_0 = I(0)$, i.e. I_0 is the irradiance at $y = 0$ (the interface) and $\alpha \equiv 2\omega n_I/c$ is called the *absorption* or (even better) the *attenuation coefficient*. The flux density will drop by a factor of $e^{-1} = 1/2.7 \approx 1/3$ after the wave has propagated a distance $y = 1/\alpha$ known as the *skin or penetration depth*. For a material to be transparent the penetration depth must be large in comparison to its thickness. The penetration depth for metals, however, is exceedingly small. For example, copper at ultraviolet wavelengths ($\lambda_0 \approx 100$ nm) has a miniscule penetration depth of about 0.6 nm while it is still roughly only 6 nm in the infrared ($\lambda_0 \approx 10{,}000$ nm). This accounts for the generally observed opacity of metals which nonetheless can become partly transparent when formed into extremely thin films (as e.g. in the case of partially silvered two-way mirrors). The familiar metallic sheen of conductors corresponds to a high reflectance which in turn arises from the fact that the incident wave cannot effectively penetrate the material. Relatively few electrons in the metal "see" the transmitted wave and therefore although each absorbs strongly, little total energy is dissipated by them. Instead, most of the incoming energy reappears as the reflected wave. By far the majority of metals including the less common ones as for example sodium, potassium, cesium, vanadium, niobium, gadolinium, holmium, yttrium, scandium, osmium and many others have a silvery-gray appearance like that of aluminum, tin or steel. They reflect almost all of the incident light regardless of wavelengths and are therefore essentially colorless.

Equation (4.77) is certainly reminiscent of Eq. (4.73) and FTIR. In both cases there is an exponential decay of the amplitude. Moreover a complete analysis would show that the transmitted waves are not strictly transverse, there being a component of the field in the direction of propagation in both instances.

The representation of metal as a continuous medium works fairly well in the low-frequency, long-wavelength domain of the infrared. Yet we certainly might expect that as the wavelength of the incident beam decreased the actual granular nature of matter would have to be reckoned with. Indeed, the continuum model shows large discrepancies

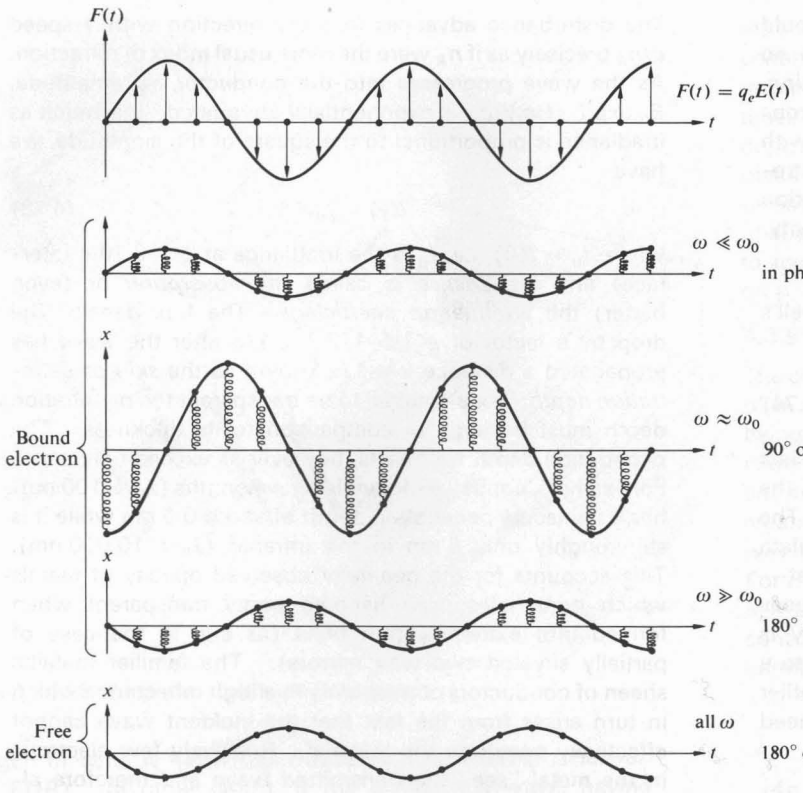


Fig. 4.34 Oscillations of bound and free electrons.

from experimental results at optical frequencies. And so we again turn to the classical atomistic picture initially formulated by Hendrik Lorentz, Paul Karl Ludwig Drude (1863–1906) and others. This simple approach will provide qualitative agreement with the experimental data but the ultimate treatment nonetheless requires quantum theory.

ii) The Dispersion Equation

Envision the conductor as an assemblage of driven, damped oscillators. Some of these correspond to free electrons and will therefore have zero restoring force while others are bound to the atom very much like those in the dielectric media of Section 3.3.1. The conduction electrons are, however, the predominant contributors to the optical properties of metals. Recall that the displacement of a vibrating electron was given (Section 3.3.1) by

$$x(t) = \frac{q_e/m_e}{(\omega_0^2 - \omega^2)} E(t).$$

With no restoring force, $\omega_0 = 0$, the displacement is opposite in sign to the driving force $q_e E(t)$ and therefore 180° out of phase with it. This is unlike the situation for transparent dielectrics where the resonance frequencies are above the visible and the electrons oscillate in phase with the driving force (Fig. 4.34). Free electrons oscillating out of phase with the incident light will reradiate wavelets which tend to cancel the incoming disturbance. The effect, as we have already seen, is a rapidly decaying refracted wave.

Assuming that the average field experienced by an electron moving about within a conductor is just the applied field $\mathbf{E}(t)$ we can extend the dispersion equation of a rare medium (3.37) to read

$$n^2(\omega) = 1 + \frac{Nq_e^2}{\epsilon_0 m_e}\left[\frac{f_e}{-\omega^2 + i\gamma_e\omega} + \sum_j \frac{f_j}{\omega_{0j}^2 - \omega^2 + i\gamma_j\omega}\right]. \quad (4.79)$$

The first bracketed term is the contribution from the free electrons wherein N is the number of atoms per unit volume. Each of these has f_e conduction electrons which have no natural frequencies. The second term arises from the bound

electrons and is identical to Eq. (3.37). It should be noted that if a metal has a particular color it is indicative of the fact that the atoms are partaking of selective absorption by way of the bound electrons. This is in addition to the general absorption characteristic of the free electrons. Recall that a medium which is very strongly absorbing at a given frequency doesn't actually absorb much of the incident light at that frequency but rather *selectively reflects* it. Gold and copper are reddish-yellow because n_I increases with wavelength and the larger values of λ are reflected more strongly. Thus, for example, gold should be fairly opaque to the longer visible wavelengths. Consequently, under white light illumination, a gold foil less than roughly 10^{-6} m thick will indeed transmit predominantly greenish-blue light.

We can get a rough idea of the response of metals to light by making a few gross simplifying assumptions. Accordingly, we neglect the bound electron contribution and furthermore assume that γ_e is also negligible for very large ω, whereupon

$$n^2(\omega) = 1 - \frac{Nq_e^2}{\epsilon_0 m_e \omega^2}. \tag{4.80}$$

The latter presumption is based on the fact that at high frequencies the electrons will undergo a great many oscillations between each collision. Free electrons and positive ions within a metal may be thought of as a plasma whose density oscillates at a natural frequency ω_p, the *plasma frequency*. This in turn can be shown to equal $(Nq_e^2/\epsilon_0 m_e)^{\frac{1}{2}}$ and so

$$n^2(\omega) = 1 - (\omega_p/\omega)^2. \tag{4.81}$$

The plasma frequency serves as a critical value below which the index is complex and the penetrating wave drops off exponentially (4.77) from the boundary; while at frequencies above ω_p, n is real, absorption is small and the conductor is transparent. In the latter circumstance n is less than one as it was for dielectrics at very high frequencies (Section 3.3.1). Hence we can expect metals in general to be fairly transparent to x-rays. Table 4.2 lists the plasma frequencies for some of the alkali metals which are transparent even to ultraviolet.

Most often the index of refraction for a metal will be complex and the impinging wave will suffer absorption in an amount which is frequency dependent. For example, the outer visors on the Apollo space suits are overlayed with a very thin film of gold (Fig. 4.35). The coating reflects about 70 % of the incident light and is used under bright conditions such as low and forward sun angles. It was designed to decrease the thermal load on the cooling system by strongly reflecting radiant energy in the infrared while still transmitting adequately in the visible. Inexpensive metal coated sunglasses which are quite similar in principle are also

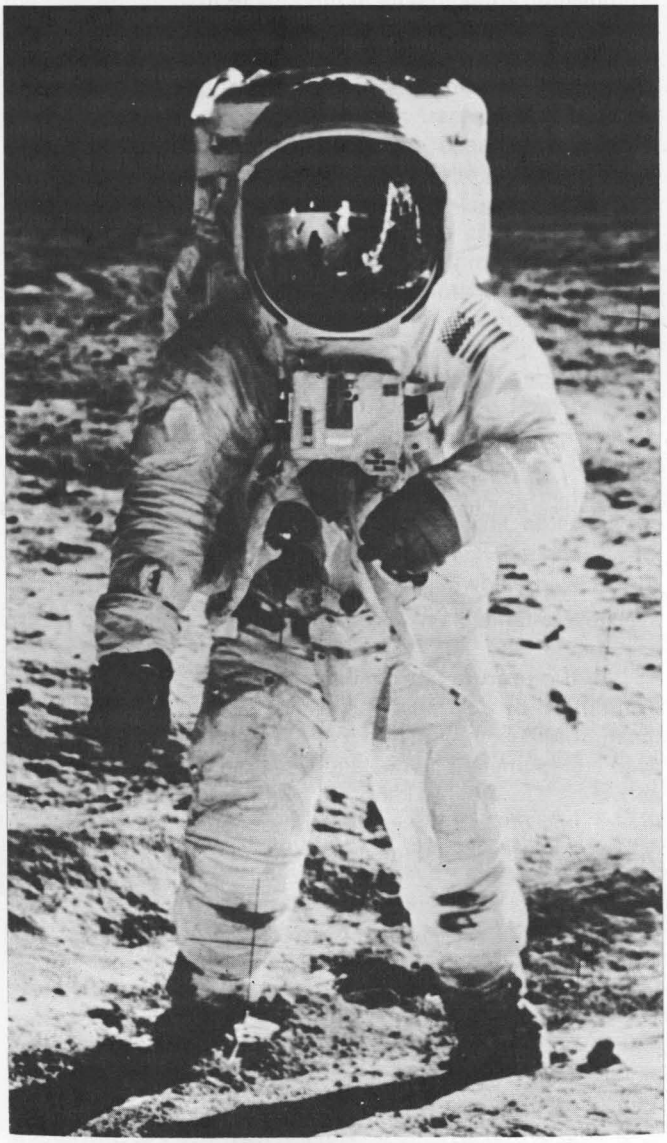


Fig. 4.35 Edwin Aldrin Jr. at Tranquility Base on the moon. The photographer, Neil Armstrong, is reflected in the gold-coated visor. [Photo courtesy NASA.]

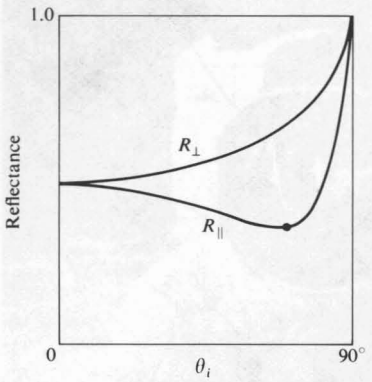


Fig. 4.36 Typical reflectance for a linearly polarized beam of white light incident on an absorbing medium.

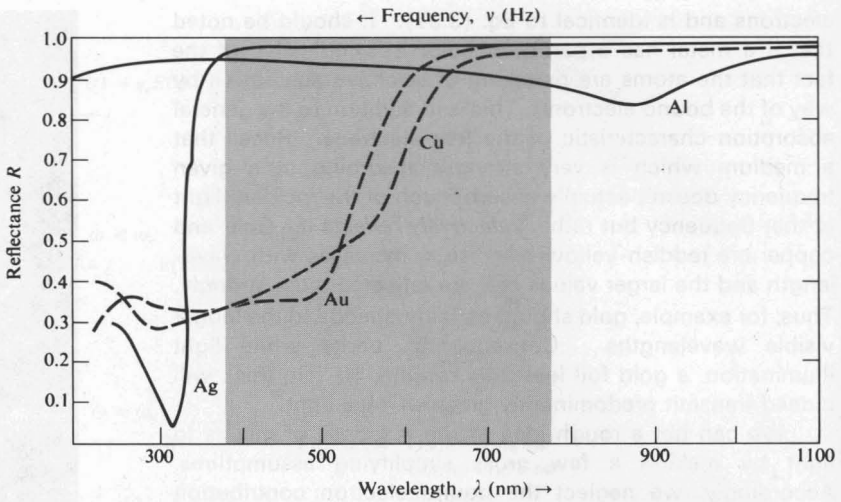


Fig. 4.37 Reflectance versus wavelength for silver, gold, copper and aluminum.

available commercially and they're well worth having just to experiment with.

The ionized upper atmosphere of the earth contains a distribution of free electrons which behave very much like those confined within a metal. The index of refraction of such a medium will be real and less than one for frequencies above ω_p. In July of 1965 the Mariner IV spacecraft made use of this effect to examine the ionosphere of the planet Mars, 216 million kilometers from earth.*

Table 4.2 Critical wavelengths and frequencies for some alkali metals.

Metal	λ_p (observed) (in nm)	λ_p (calculated) (in nm)	$\nu_p = c/\lambda_p$ (observed) (in Hz)
Lithium Li	155	155	1.94×10^{15}
Sodium Na	210	209	1.43×10^{15}
Potassium K	315	287	0.95×10^{15}
Rubidium Rb	340	322	0.88×10^{15}

* R. Von Eshelman, *Sci. Am.* **220**, 78 (1969).

If we wish to communicate between two distant terrestrial points we might bounce low-frequency waves off the earth's ionosphere. In contrast, in order to speak to someone on the moon we should use high-frequency signals to which the ionosphere would be transparent.

iii) Reflection from a Metal

Imagine that a plane wave initially in air impinges on a conducting surface. The transmitted wave advancing at some angle to the normal will be inhomogeneous. But if the conductivity of the medium is increased, the wavefronts will become aligned with the surfaces of constant amplitude, whereupon $\mathbf{k}_t$ and $\hat{\mathbf{u}}_n$ would approach parallelism. In other words, in a good conductor the transmitted wave propagates in a direction normal to the interface regardless of θ_i.

Let's now compute the reflectance, $R = I_r/I_i$, for the simplest case of normal incidence on a metal. Taking $n_i = 1$ and $n_t = n_c$ (i.e. the complex index) we have, from Eq. (4.47), that

$$R = \left(\frac{n_c - 1}{n_c + 1}\right)\left(\frac{n_c - 1}{n_c + 1}\right)^* \tag{4.82}$$

and therefore since $n_c = n_R - in_I$

$$R = \frac{(n_R - 1)^2 + n_I^2}{(n_R + 1)^2 + n_I^2}. \tag{4.83}$$

If the conductivity of the material goes to zero we have

the case of a dielectric whereupon in principle the index is real ($n_I = 0$) and the attenuation coefficient, α, is zero. Under those circumstances, the index of the transmitting medium n_t is n_R and the reflectance (4.83) becomes identical with that of Eq. (4.67). If instead n_I is large while n_R is comparatively small R in turn becomes large (Problem 4.40). In the unattainable limit where n_c is purely imaginary, one hundred percent of the incident flux density would be reflected ($R = 1$). Notice that it is possible for the reflectance of one metal to be greater than that of another even though its n_I is smaller. For example at $\lambda_0 = 589.3$ nm the parameters associated with solid sodium are roughly $n_R = 0.04$, $n_I = 2.4$, and $R = 0.9$; those for bulk tin are $n_R = 1.5$, $n_I = 5.3$, and $R = 0.8$; while for a gallium single crystal $n_R = 3.7$, $n_I = 5.4$, and $R = 0.7$.

The curves of $R_{\parallel}$ and $R_{\perp}$ for oblique incidence shown in Fig. 4.36 are somewhat typical of absorbing media. Thus while R at $\theta_i = 0$ for gold is about 0.5, as opposed to near 0.9 for silver in white light, both have reflectances which are quite similar in shape approaching 1.0 at $\theta_i = 90°$. Just as with dielectrics (Fig. 4.28) $R_{\parallel}$ drops to a minimum at what is now called the *principal angle of incidence* but here that minimum is nonzero. Figure 4.37 illustrates the spectral reflectance at normal incidence for a number of evaporated metal films under ideal conditions. Observe that while gold transmits fairly well in and below the green region of the spectrum, silver which is highly reflective across the visible becomes transparent in the ultraviolet at about 316 nm.

Phase shifts arising from reflection off a metal occur in both components of the field (i.e. parallel and perpendicular to the plane of incidence). These are generally neither 0 nor π with a notable exception at $\theta_i = 90°$ where, just as with a dielectric, both components shift phase by 180° on reflection.

4.4 FAMILIAR ASPECTS OF THE INTERACTION OF LIGHT AND MATTER

Let's now examine some of the phenomena which paint the everyday world in a marvel of myriad colors.

The mechanism responsible for the yellowish-red hue of gold and copper is, in some respects, similar to the process which causes the sky to appear blue. Putting it rather succinctly (see Section 8.5 for a further discussion of scattering in the atmosphere), the molecules of air have resonances in the ultraviolet and will therefore be driven into larger amplitude oscillations as the frequency of the incident light increases toward the ultraviolet. Consequently they will effectively take energy from and reemit (i.e. scatter) the blue component of sunlight in all directions, transmitting the complementary red end of the spectrum with little alteration. This is in analogy with the selective reflection or scattering of yellow-red light that takes place at the surface of a gold film and the concomitant transmission of blue-green light. In contradistinction, the characteristic colors of most substances have their origin in the phenomenon of *selective or preferential absorption*. For example, water has a very light green-blue tint by dint of its absorption of red light. That is, the H_2O molecules have a broad resonance in the infrared which extends somewhat into the visible. The absorption isn't very strong so that there is no accentuated reflection of red light at the surface. Rather it is transmitted and gradually absorbed out until at a depth of about 30 m of sea water, red is almost completely removed from sunlight. This same process of selective absorption is responsible for the colors of brown eyes and butterflies, of birds and bees and cabbages and kings. Indeed the great majority of objects in nature appear to have characteristic colors as the result of preferential absorption by pigment molecules. In contrast with most atoms and molecules, which have resonances in the ultraviolet and infrared, the pigment molecules must obviously have resonances in the visible. Yet visible photons have energies of roughly 1.6 eV to 3.2 eV which, as you might expect, are on the low side for ordinary electron excitation and on the high side for excitation via molecular vibration. Despite this, there are atoms where the bound electrons form incomplete shells (e.g. gold) and variations in the configuration of these shells provide a mode for low-energy excitation. In addition to these there is the large group of organic dye molecules which evidently also have resonances in the visible. All such substances, whether natural or synthetic, consist of long chain molecules made up of regularly alternating single and double bonds in what is called a conjugated system. This structure is typified by the carotene molecule $C_{40}H_{56}$ (Fig. 4.38). The carotenoids range in color from yellow to red and are found in carrots, tomatoes, daffodils, dandelions, autumn leaves and people. The chlorophylls are another group of familiar natural pigments but here a portion of the long chain is turned around on itself to form a ring. In any event, conjugated systems of this sort contain a number of particularly mobile electrons known as *pi electrons*. These are not bound to specific atomic sites but instead can range over the relatively large dimensions of the molecular chain or ring. In the phraseology of quantum mechanics we would say that these were long wavelength, low frequency, and therefore low energy, electron states. The energy required to raise a pi electron to an excited state is

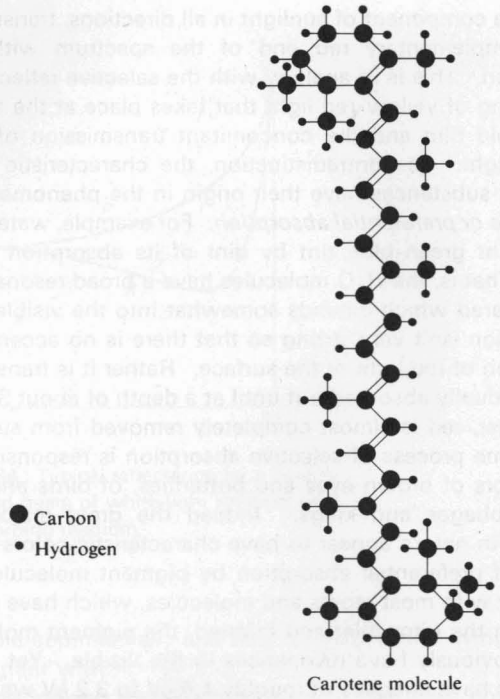


Fig. 4.38 The carotene molecule.

accordingly comparatively low corresponding to that of visible photons. In effect, we can imagine the molecule as an oscillator having a resonance frequency in the visible.

The energy levels of an individual atom are precisely defined i.e. the resonances are very sharp. With solids and liquids however the proximity of the atoms results in a broadening of the energy levels into wide bands. In other words, the resonances spread over a broad range of frequencies. Consequently we can expect that a dye will not absorb just a narrow portion of the spectrum; indeed if it did it would reflect most frequencies and appear nearly white.

To see how all of these ideas fit together imagine that we have a piece of blue stained glass or cellophane. White light impinging on the surface (Fig. 4.39) is partially reflected, the remainder being transmitted into the medium. As the beam advances, more and more of the red end of the spectrum is removed from it by absorption until it finally emerges as blue light. It should be clear from the figure that the colors of stained glass windows will be far more vivid when viewed in transmitted rather than reflected light. It is precisely this penetration and gradual preferential absorption which is responsible for the colors of most objects. For example, the fibers of a sample of white cloth or paper are essentially transparent but when dyed each fiber behaves as if it were a chip of colored glass. The incident light penetrates the paper emerging for the most part as a reflected beam only after undergoing numerous reflections and refractions within the dyed fibers. The exiting light will be colored to the extent that it is minus that frequency component which is absorbed by the dye. This is precisely why a leaf appears green or a banana yellow.

A bottle of ordinary blue ink looks blue in either reflected or transmitted light. But if we paint it on a glass slide and evaporate out the solvent, something rather interesting happens. The concentrated pigment absorbs so effectively that it preferentially reflects at the resonant frequency and we are back to the idea that a strong absorber (large n_I) is a strong reflector. Thus, concentrated bluish ink reflects magenta and transmits blue, while red ink will reflect green and transmit red. Try it with a felt marking pen but you must use reflected light, being careful not to inadvertently inundate the sample with unwanted light from below. (Wipe the ink into a thin layer and then place the slide on a piece of black paper.)

If the range of frequencies being absorbed spreads across the visible, the object will appear black. That is not to say that there is no reflection at all—you obviously can see a reflected image in a piece of black patent leather and a rough black surface reflects as well, only diffusely. If you still have

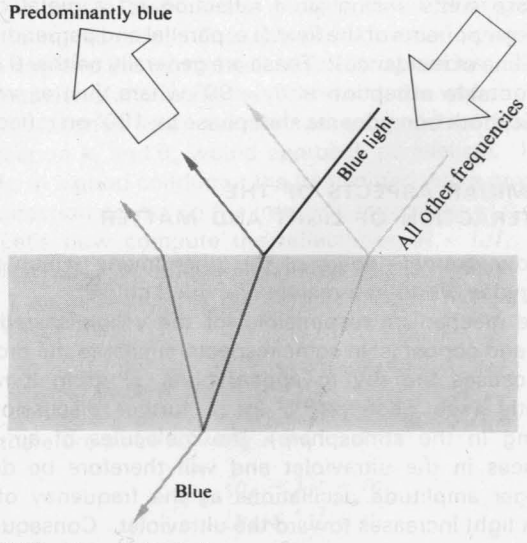


Fig. 4.39 Blue stained glass.

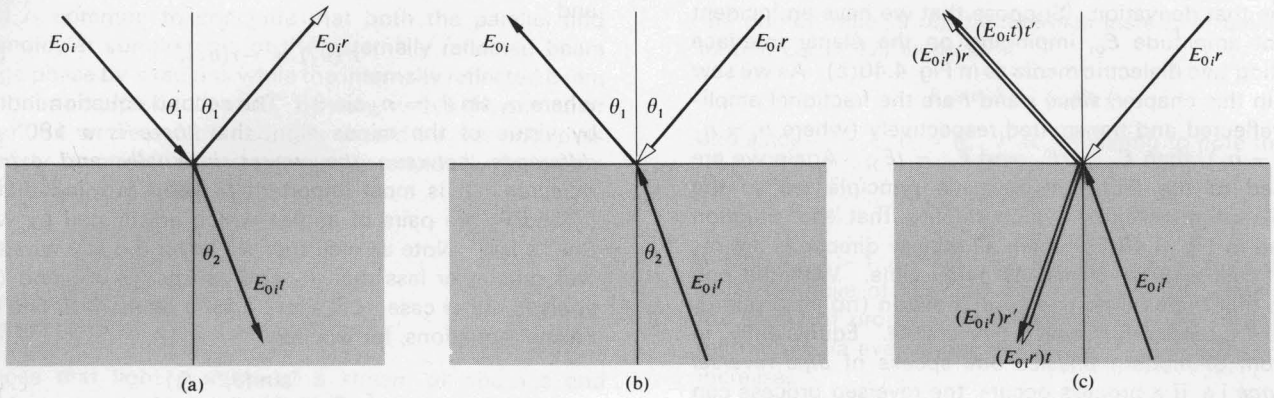


Fig. 4.40 Reflection and refraction via the Stokes treatment.

those red and blue inks, mix them, add some green, and you'll get black.

At the other extreme an object is white under white light illumination when it reflects both effectively and uniformly over the visible spectrum. Although water is essentially transparent, water vapor appears white as does ground glass. The reason is simple enough—if the grain size is small but much larger than the wavelengths involved, light will enter each transparent particle, be reflected and refracted several times and emerge. There will be no distinction made between any of the frequency components and so the reflected light reaching the observer will be white. This is the mechanism accountable for the whiteness of things like sugar, salt, paper, clouds, talcum powder, snow and paint, each grain of which is actually transparent. Similarly clear plastic tape on a thick roll appears to be opaque as does an ordinarily transparent material filled with small air bubbles (e.g. beaten egg white). Even though we usually think of paper, talcum powder, sugar, etc. as each consisting of some sort of opaque white substance, it's an easy matter to dispel that misconception. Cover a printed page with a few of these materials (a sheet of white paper, some grains of sugar or talcum) and illuminate it from behind. You'll have little difficulty in seeing through them. In the case of white paint, one simply suspends colorless transparent particles such as the oxides of zinc, titanium or lead in an equally transparent vehicle, as for example, linseed oil or the newer acrylics. Obviously if the particles and vehicle have the same index of refraction there will not be any reflections at the grain boundaries. The particles will simply disappear into the conglomeration which itself remains clear. In contrast, if the indices are markedly different there will be a good deal of reflection at all wavelengths (Problem 4.33) and the paint will appear white and opaque [take another look at Eq. (4.67)]. To color paint one need only dye the particles so that they absorb out all frequencies except the desired range.

Carrying the logic in the reverse direction, if we reduce the relative index, n_{ti}, at the grain or fiber boundaries, the particles of material will reflect less, thereby decreasing the overall whiteness of an object. Consequently, a wet white tissue will have a grayish more transparent look. Wet talcum powder loses its sparkling white becoming a dull gray as does wet white cloth. And in the same way a piece of dyed fabric soaked in a clear liquid (e.g. water, gin or benzene) will lose its whitish haze and become very much darker, the colors then being deep and rich like those of a still wet water color painting.

In addition to the above processes that are specifically related to reflection, refraction and absorption, there are many other color-generating mechanisms which we shall explore later on. For example, the scarabeid beetles mantle themselves in the brilliant colors produced by diffraction gratings grown on their wing-cases; while wavelength-dependent interference effects contribute to the color patterns seen on oil slicks, mother-of-pearl, soap bubbles, peacocks and hummingbirds.

4.5 THE STOKES TREATMENT OF REFLECTION AND REFRACTION

A rather elegant and novel way of looking at reflection and transmission at a boundary was developed by the British physicist Sir George Gabriel Stokes (1819–1903). Since we will often make use of his results in the future let's now

examine that derivation. Suppose that we have an incident wave of amplitude E_{0i} impinging on the planar interface separating two dielectric media as in Fig. 4.40(a). As we saw earlier in this chapter, since r and t are the fractional amplitudes reflected and transmitted respectively (where $n_i = n_1$ and $n_t = n_2$), then $E_{0r} = rE_{0i}$ and $E_{0t} = tE_{0i}$. Again we are reminded of the fact that Fermat's principle led to the principle of reversibility which implies that the situation depicted in Fig. 4.40(b) where all the ray directions are reversed must also be physically permissible. With the one proviso that there be no energy dissipation (no absorption), a wave's meanderings must be reversible. Equivalently, in the idiom of modern physics one speaks of *time-reversal invariance* i.e. if a process occurs, the reversed process can also occur. Thus if we take a hypothetical motion picture of the wave incident on, reflecting from and transmitting through the interface, the behavior depicted when the film is run backwards must also be physically realizable. Accordingly, examine Fig. 4.40(c) where there are now two incident waves of amplitudes $E_{0i}r$ and $E_{0i}t$. A portion of the wave whose amplitude is $E_{0i}t$ is both reflected and transmitted at the interface. Without making any assumptions let r' and t' be the amplitude reflection and transmission coefficients for a wave incident from below (i.e. $n_i = n_2$, $n_t = n_1$). Consequently, the reflected portion is $E_{0i}tr'$ while that transmitted is $E_{0i}tt'$. Similarly the incoming wave whose amplitude is $E_{0i}r$ splits into segments of amplitude $E_{0i}rr$ and $E_{0i}rt$. If the configuration of Fig. 4.40(c) is to be identical with that of Fig. 4.40(b) then obviously

$$E_{0i}tt' + E_{0i}rr = E_{0i} \tag{4.84}$$

and

$$E_{0i}rt + E_{0i}tr' = 0. \tag{4.85}$$

Hence

$$tt' = 1 - r^2 \tag{4.86}$$

and

$$r' = -r, \tag{4.87}$$

the latter two equations being known as the Stokes relations. Actually this discussion calls for a bit more caution than is most often granted it. It must be pointed out that *the amplitude coefficients are functions of the incident angles* and therefore Stokes' relations might better be written as

$$t(\theta_1)t'(\theta_2) = 1 - r^2(\theta_1) \tag{4.88}$$

and

$$r'(\theta_2) = -r(\theta_1), \tag{4.89}$$

where $n_1 \sin\theta_1 = n_2 \sin\theta_2$. The second equation indicates, by virtue of the minus sign, that *there is a* 180° *phase difference between the waves internally and externally reflected.* It is most important to keep in mind that here θ_1 and θ_2 are pairs of angles which are related by way of Snell's law. Note as well that we never did say whether n_1 was greater or less than n_2 and so Eqs. (4.88) and (4.89) apply in either case. Let's return for a moment to one of the Fresnel equations, for example

$$r_\perp = -\frac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)}. \tag{4.42}$$

If a ray enters from above as in Fig. 4.40(a) and we assume $n_2 > n_1$, $r_\perp$ is computed by setting $\theta_i = \theta_1$ and $\theta_t = \theta_2$ (external reflection) the latter arrived at from Snell's law. If on the other hand the wave is incident at that same angle from below (in this instance internal reflection) $\theta_i = \theta_1$ and we again substitute in Eq. (4.42) but here θ_t is not θ_2 as before. The values of $r_\perp$ for internal and external reflection *at the same incident angle* are obviously different. Now then suppose, in this case of internal reflection, that we take $\theta_i = \theta_2$. Then $\theta_t = \theta_1$, the ray directions are the reverse of those in the first situation and Eq. (4.42) yields

$$r'_\perp(\theta_2) = -\frac{\sin(\theta_2 - \theta_1)}{\sin(\theta_2 + \theta_1)}.$$

Although it may be unnecessary we once again point out that this is just the negative of what was determined for $\theta_i = \theta_1$ and external reflection i.e.

$$r'_\perp(\theta_2) = -r_\perp(\theta_1). \tag{4.90}$$

The use of primed and unprimed symbols to denote the amplitude coefficients should serve as a reminder that we are once more dealing with angles related by Snell's law. In the same way interchanging θ_i and θ_t in Eq. (4.43) leads to

$$r'_{\parallel}(\theta_2) = -r_{\parallel}(\theta_1). \tag{4.91}$$

The 180° phase difference between each pair of components is evident in Fig. 4.23 but do keep in mind that when $\theta_i = \theta_p$, $\theta_t = \theta'_p$ and vice versa (Problem 4.37). Beyond $\theta_i = \theta_c$ there is no transmitted wave, Eq. (4.89) is not applicable and, as we have seen, the phase difference is no longer 180°.

It is common to conclude that both the parallel and perpendicular components of the externally reflected beam change phase by π radians while the internally reflected beam undergoes no phase shift at all. By now, within the particular convention we've established, this should be recognized as obviously incorrect, or at least almost obviously [compare Figs. 4.24(a) and 4.25(a)].

4.6 PHOTONS AND THE LAWS OF REFLECTION AND REFRACTION

Suppose that light consists of a stream of photons and consider one such photon which strikes the interface between two dielectric media at an angle θ_i and is subsequently transmitted across it at an angle θ_t. We know that if this were just one out of billions of such quanta in a narrow laser beam it would obediently conform to Snell's law. To appreciate this behavior let's examine the dynamics associated with the odyssey of our single photon. Recall that

$$\mathbf{p} = \hbar\mathbf{k} \qquad [3.60]$$

and consequently the incident and transmitted momenta are $\mathbf{p}_i = \hbar\mathbf{k}_i$ and $\mathbf{p}_t = \hbar\mathbf{k}_t$ respectively. We assume (without much justification) that while the material in the vicinity of the interface affects the y-component of momentum it leaves the x-component unchanged. Indeed we know experimentally that linear momentum can be transferred to a medium from a light beam (Section 3.4.2). The statement of conservation of the component of momentum parallel to the interface takes the form

$$p_{ix} = p_{tx} \qquad (4.92)$$

or

$$p_i \sin\theta_i = p_t \sin\theta_t.$$

If we use Eq. (3.60) this becomes

$$k_i \sin\theta_i = k_t \sin\theta_t$$

and hence

$$\frac{1}{\lambda_i}\sin\theta_i = \frac{1}{\lambda_t}\sin\theta_t.$$

Multiplying both sides by c/v we have

$$n_i \sin\theta_i = n_t \sin\theta_t,$$

which of course is Snell's law. In exactly the same way if the photon reflects off the interface instead of being transmitted, Eq. (4.92) leads to

$$k_i \sin\theta_i = k_r \sin\theta_r$$

and since $\lambda_i = \lambda_r$, $\theta_i = \theta_r$. It is interesting to note that

$$n_{ti} = \frac{p_t}{p_i} \qquad (4.93)$$

and so if $n_{ti} > 1$, $p_t > p_i$. Experiments dating back as far as 1850 to those of Foucault have shown that when $n_{ti} > 1$ the speed of propagation is actually reduced in the transmitting media even though here the momentum apparently increases.*

Do keep in mind that we have been dealing with a very simple representation which leaves much to be desired, for example it says nothing about the atomic structure of the media and nothing about the probability or likelihood of a photon traversing a given path. Even though this treatment is obviously simplistic, it is appealing pedagogically (see Chapter 13).

PROBLEMS

4.1 Calculate the transmission angle for a ray incident in air at 30° on a block of crown-glass ($n_g = 1.52$).

4.2* A ray of yellow light from a sodium discharge lamp falls on the surface of a diamond in air at 45°. If at that frequency $n_d = 2.42$, compute the angular deviation suffered upon transmission.

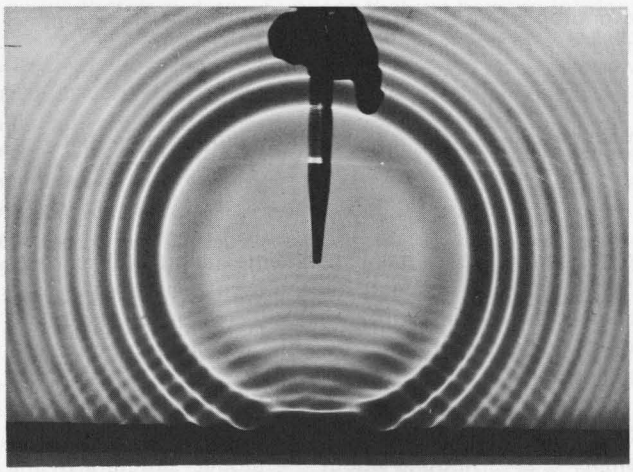
4.3 Use Huygens' construction to create a wavefront diagram showing the form a spherical wave will have after reflection from a planar surface as in the ripple tank photos of Fig. 4.41. Draw the ray diagram as well.

4.4* Given a water ($n_w = 1.33$) glass ($n_g = 1.50$) interface, compute the transmission angle for a beam incident in the water at 45°. If the transmitted beam is reversed so that it now impinges on the interface, show that $\theta_t = 45°$.

4.5 A beam of 12 cm planar microwaves strikes the surface of a dielectric at 45°. If $n_{ti} = \frac{4}{3}$, compute (a) the wavelength in the transmitting medium and (b) the angle θ_t.

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* This suggests an increase in the photon's effective mass. See F. R. Tangherlini, "On Snell's Law and the Gravitational Deflection of Light," *Am. J. Phys.* **36**, 1001 (1968). Take a *cautious* look at R. A. Houstoun, "Nature of Light," *J. Opt. Soc. Am.* **55**, 1186 (1965).



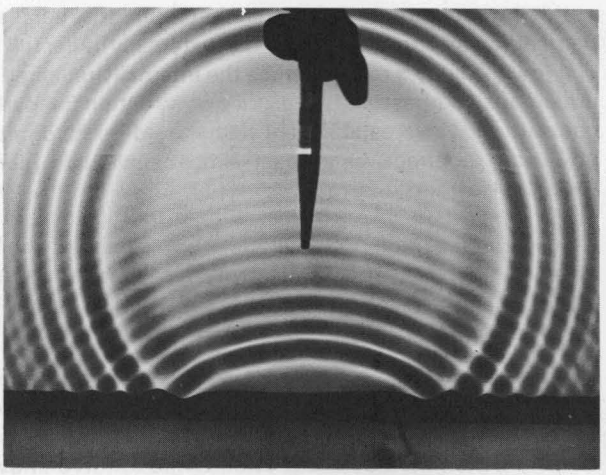


Fig. 4.41 [Photo courtesy *Physics*, Boston, D. C. Heath & Co., 1960.]

4.6* Light of wavelength 600 nm in vacuum enters a block of glass where $n_g = 1.5$. Compute its wavelength in the glass. What color will it appear to someone imbedded in the glass (see Table 3.2)?

4.7 Figure 4.42 shows a bundle of rays entering and emerging from a glass disk (a lens). From the configuration of the rays, determine the shape of the wavefronts at various points. Draw a diagram in profile.

4.8 Make a plot of θ_i versus θ_t for an air–glass boundary where $n_{ga} = 1.5$.

4.9 Making use of the ideas of equal transit times between corresponding points and the orthogonality of rays and wavefronts, derive the law of reflection and Snell's law. The ray diagram of Fig. 4.43 should be helpful.

4.10 Starting with Snell's law prove that the vector refraction equation has the form

$$n_t\hat{\mathbf{k}}_t - n_i\hat{\mathbf{k}}_i = (n_t\cos\theta_t - n_i\cos\theta_i)\hat{\mathbf{u}}_n. \qquad [4.8]$$

4.11 Derive a vector expression equivalent to the law of reflection. As before, let the normal go from the incident to the transmitting medium even though it obviously doesn't really matter.

4.12 In the case of reflection from a planar surface use Fermat's principle to prove that the incident and reflected rays share a common plane with the normal $\hat{\mathbf{u}}_n$, namely the plane of incidence.

4.13* Derive the law of reflection, $\theta_i = \theta_r$, by using the calculus to minimize the transit time as required by Fermat's principle.

4.14 Show analytically that a beam entering a planar transparent plate, as in Fig. 4.44, emerges parallel to its initial direction. Derive an expression for the lateral displacement of the beam. Incidentally, the fact that the incoming and outgoing rays are parallel would be true even for a stack of plates of different material.

4.15* Show that the two rays which enter the system in Fig. 4.45 parallel to each other emerge from it being parallel.

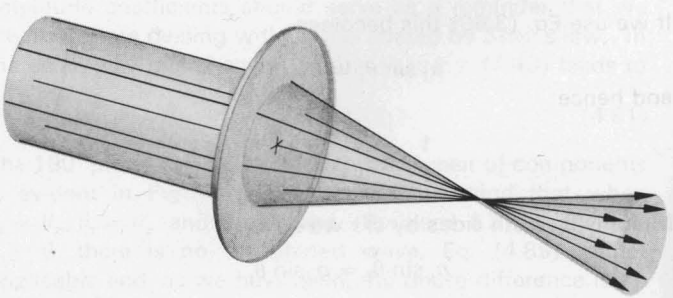


Fig. 4.42

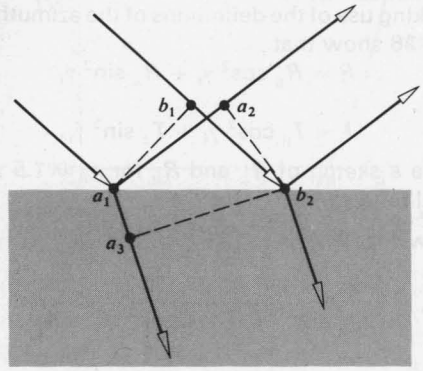


Fig. 4.43

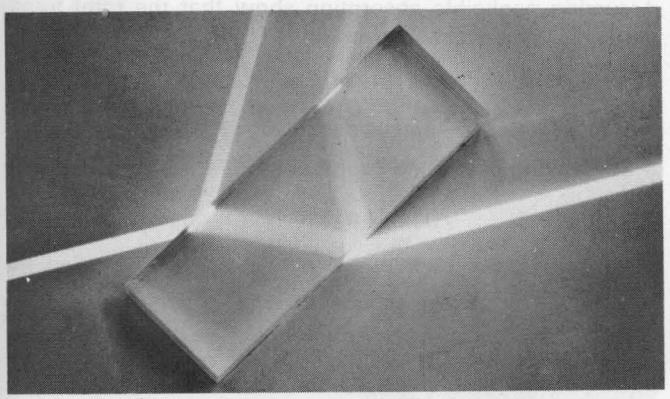


Fig. 4.44

4.16 Discuss the results of Problem 4.14 in the light of Fermat's principle, i.e. how does the relative index n_{21} affect things? To see the lateral displacement look at a broad source through a thick piece of glass ($\approx \frac{1}{4}$ inch) or a stack (four will do) of microscope slides *held at an angle*. There will be an obvious shift between the region of the source seen directly and the region viewed through the glass.

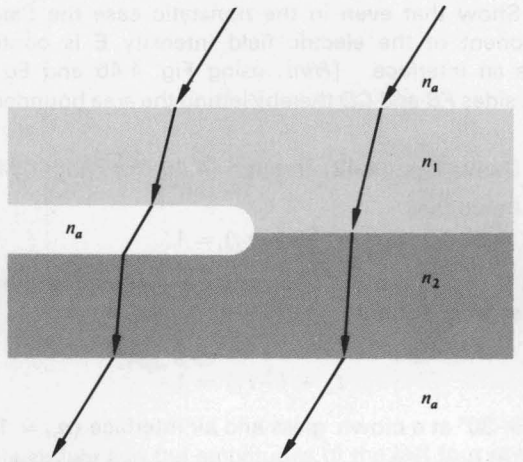


Fig. 4.45

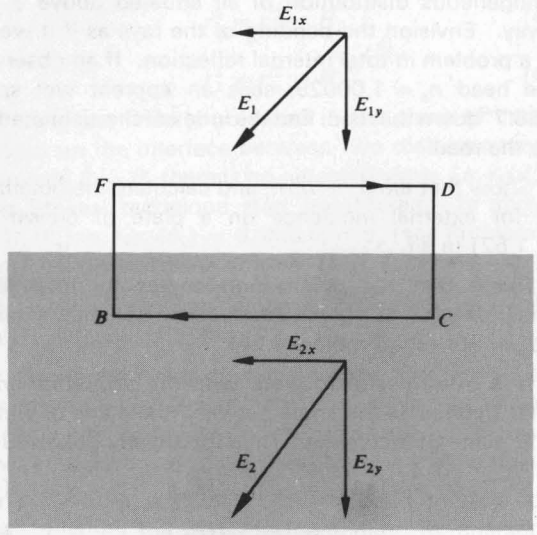


Fig. 4.46

4.17 Suppose that a light wave which is linearly polarized in the plane of incidence impinges at 30° on a crown glass ($n_g = 1.52$) plate in air. Compute the appropriate amplitude reflection and transmission coefficients at the interface. Compare your results with Fig. 4.20.

4.18 Show that even in the nonstatic case the tangential component of the electric field intensity **E** is continuous across an interface. [*Hint*: using Fig. 4.46 and Eq. (3.5) shrink sides *FB* and *CD* thereby letting the area bounded go to zero.]

4.19 Derive Eqs. (4.42) through (4.45) for $r_\perp$, $r_{||}$, $t_\perp$, and $t_{||}$.

4.20 Prove that

$$t_\perp + (-r_\perp) = 1 \qquad [4.49]$$

for all θ_i, first from the boundary conditions and then from the Fresnel equations.

4.21* Verify that

$$t_\perp + (-r_\perp) = 1 \qquad [4.49]$$

for $\theta_i = 30°$ at a crown glass and air interface ($n_{ti} = 1.52$).

4.22* Calculate the critical angle beyond which there is total internal reflection at an air–glass ($n_g = 1.5$) interface. Compare this result with that of Problem 4.8.

4.23 Consider the common mirage associated with an inhomogeneous distribution of air situated above a warm roadway. Envision the bending of the rays as if it were instead a problem in total internal reflection. If an observer, at whose head $n_a = 1.00029$, sees an apprent wet spot at $\theta_i \geq 88.7°$ down the road, find the index of the air immediately above the road.

4.24 Show that $\tan \theta_p = n_t/n_i$ and calculate the polarization angle for external incidence on a plate of crown glass ($n_g = 1.52$) in air.

4.25 Show that the polarization angles for internal and external reflection at a given interface are complementary, i.e. $\theta_p + \theta'_p = 90°$ (see Problem 4.24).

4.26 It is often useful to work with the *azimuthal angle* γ which is defined as the angle between the plane of vibration and the plane of incidence. Thus for linearly polarized light

$$\tan \gamma_i = [E_{0i}]_\perp / [E_{0i}]_{||} \qquad (4.94)$$

$$\tan \gamma_t = [E_{0t}]_\perp / [E_{0t}]_{||} \qquad (4.95)$$

and

$$\tan \gamma_r = [E_{0r}]_\perp / [E_{0r}]_{||}. \qquad (4.96)$$

Figure 4.47 is a plot of γ_r versus θ_i for internal and external reflection at an air–glass interface ($n_{ga} = 1.51$), where $\gamma_i = 45°$. Verify a few of the points on the curves and in addition show that

$$\tan \gamma_r = -\frac{\cos(\theta_i - \theta_t)}{\cos(\theta_i + \theta_t)} \tan \gamma_i. \qquad (4.97)$$

4.27* Making use of the definitions of the azimuthal angles in Problem 4.26 show that

$$R = R_{||} \cos^2 \gamma_i + R_\perp \sin^2 \gamma_i \qquad (4.98)$$

and

$$T = T_{||} \cos^2 \gamma_i + T_\perp \sin^2 \gamma_i. \qquad (4.99)$$

4.28 Make a sketch of $R_\perp$ and $R_{||}$ for $n_i = 1.5$ and $n_t = 1$, i.e. internal reflection.

4.29 Show that

$$T_{||} = \frac{\sin 2\theta_i \sin 2\theta_t}{\sin^2(\theta_i + \theta_t) \cos^2(\theta_i - \theta_t)} \qquad (4.100)$$

and

$$T_\perp = \frac{\sin 2\theta_i \sin 2\theta_t}{\sin^2(\theta_i + \theta_t)}. \qquad (4.101)$$

4.30* Using the results of Problem 4.29, i.e. Eqs. (4.100) and (4.101), show that

$$R_{||} + T_{||} = 1 \qquad [4.65]$$

and

$$R_\perp + T_\perp = 1. \qquad [4.66]$$

4.31 Suppose that we look at a source perpendicularly through a stack of N microscope slides. The source seen through even a dozen slides will be noticeably darker. Assuming negligible absorption show that the total transmittance of the stack is given by

$$T_t = (1 - R)^{2N}$$

and evaluate T_t for three slides in air.

4.32 Making use of the expression

$$I(y) = I_0 e^{-\alpha y} \qquad [4.78]$$

for an absorbing medium we define a quantity called the *unit transmittance* T_1. At normal incidence (4.55) $T = I_t/I_i$ and thus when $y = 1$, $T_1 \equiv I(1)/I_0$. If the total thickness of the slides in the previous problem is d and if they now have a transmittance per unit length T_1, show that

$$T_t = (1 - R)^{2N} (T_1)^d.$$

4.33 Show that at normal incidence on the boundary between two dielectrics as $n_{ti} \to 1$, $R \to 0$, and $T \to 1$. Moreover, prove that as $n_{ti} \to 1$, $R_{||} \to 0$, $R_\perp \to 0$, $T_{||} \to 1$, and $T_\perp \to 1$ for all θ_i. Thus as the two media take on more similar indices of refraction, less and less energy is carried off in the reflected wave. It should be obvious that when $n_{ti} = 1$ there will be no interface and no reflection.

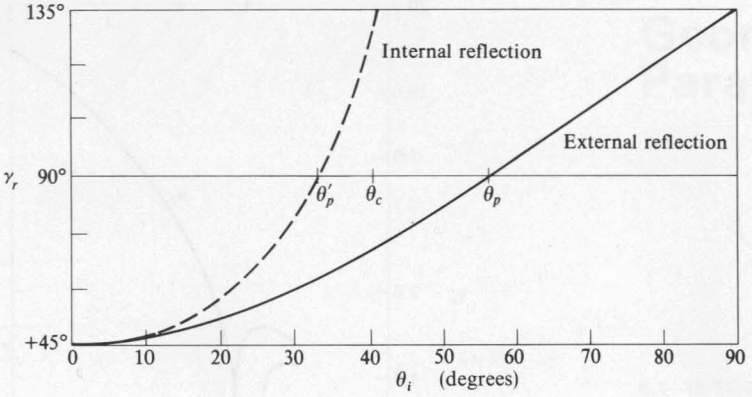


Fig. 4.47

4.34* Derive the expressions for $r_\perp$ and $r_{||}$ given by Eqs. (4.70) and (4.71).

4.35 Show that when $\theta_i > \theta_c$ at a dielectric interface, $r_{||}$ and $r_\perp$ are complex and $r_\perp r_\perp^* = r_{||} r_{||}^* = 1$.

4.36 Figure 4.48 depicts a ray being multiply reflected by a transparent dielectric plate (the amplitudes of the resulting fragments are indicated). As in Section 4.5 we use the primed coefficient notation because the angles are related by Snell's law.

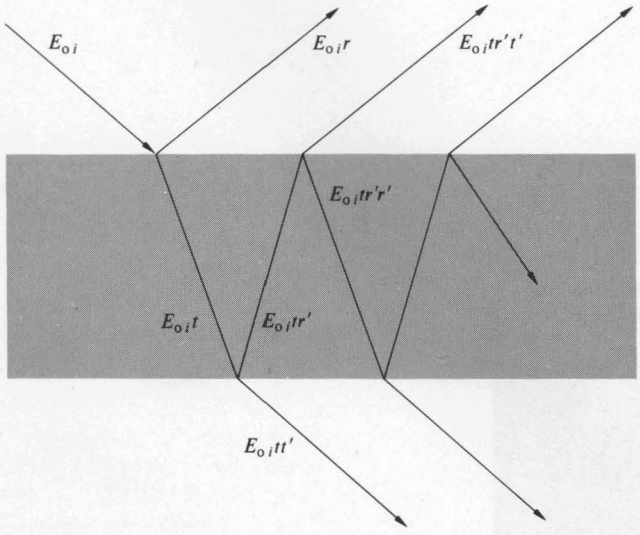


Fig. 4.48

a) Finish labeling the amplitudes of the last four rays.
b) Show, using the Fresnel equations, that

$$t_{||}t'_{||} = T_{||} \tag{4.102}$$

$$t_\perp t'_\perp = T_\perp \tag{4.103}$$

$$r_{||}^2 = r_{||}'^2 = R_{||} \tag{4.104}$$

and

$$r_\perp^2 = r_\perp'^2 = R_\perp. \tag{4.105}$$

4.37* A wave, linearly polarized in the plane of incidence, impinges on the interface between two dielectric media. If $n_i > n_t$ and $\theta_i = \theta'_p$ there is no reflected wave, i.e. $r'_{||}(\theta'_p) = 0$. Using Stokes' technique start from scratch to show that $t_{||}(\theta_p)t'_{||}(\theta'_p) = 1$, $r_{||}(\theta_p) = 0$, and $\theta_t = \theta_p$ (see Problem 4.25). How does this compare with Eq. (4.102)?

4.38 Use the Fresnel equations to show that $t_{||}(\theta_p)t'_{||}(\theta'_p) = 1$ as in the previous problem.

4.39 Figure 4.49 depicts a glass cube surrounded by four glass prisms which are in very close proximity to its sides. Sketch in the paths which will be taken by the two rays shown and discuss a possible application for the device.

4.40 Figure 4.50 is a plot of n_I and n_R versus λ for a common metal. Identify the metal by comparing its characteristics with those considered in the chapter and discuss its optical properties.

4.41 Figure 4.51 shows a prism-coupler arrangement developed at the Bell Telephone Laboratories. Its function is to feed a laser beam into a thin (0.00001 in) transparent film which then serves as a sort of waveguide. One possible application is that of thin film laser beam circuitry—a kind of integrated optics. How do you think the thing works?

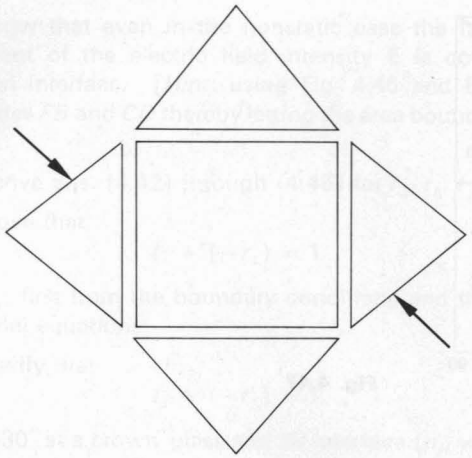


Fig. 4.49

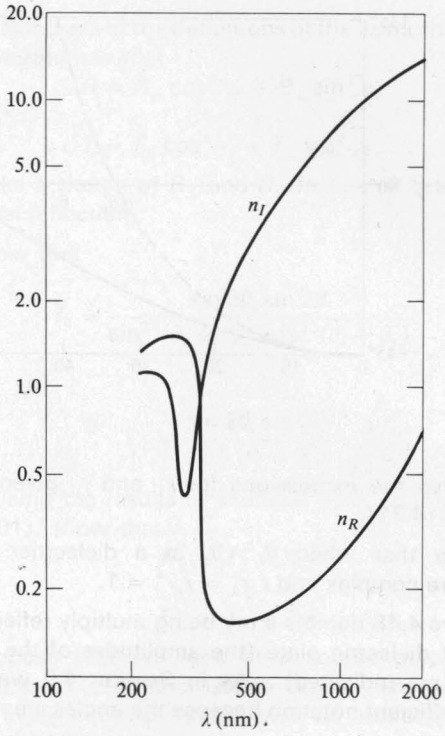


Fig. 4.50

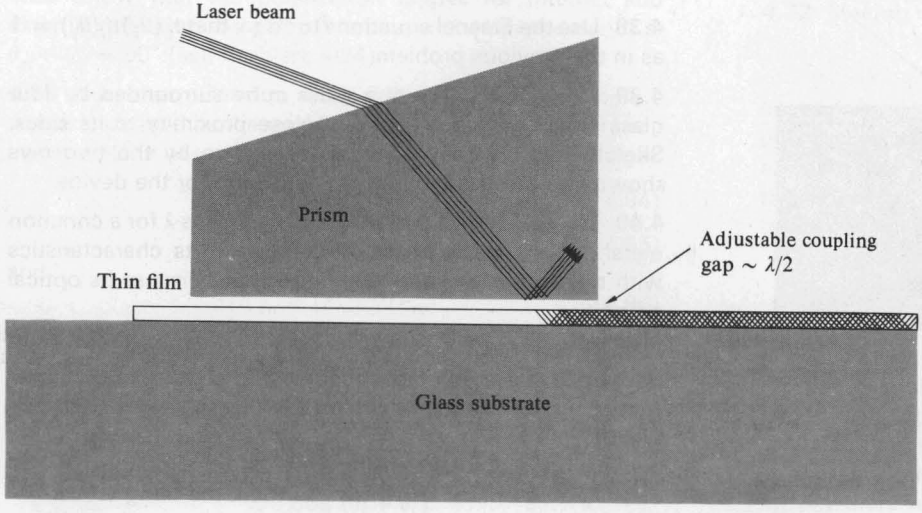


Fig. 4.51